Understanding Multilateral Tariffs and Exchange Rates*

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Abstract

The standard prediction that an import tariff appreciates the tariffing country's currency comes from a two-country setting, where the bilateral exchange rate, effective exchange rate, and the terms of trade coincide. In a multilateral setting, these measures separate and respond differently to tariffs. Under symmetric trade relationships, we show that a unilateral tariff appreciates the effective exchange rate but can depreciate the home currency against untariffed bystanders when substitution across foreign suppliers is sufficiently elastic. Tariff retaliation can instead lead both counterparts to depreciate in effective terms. These results are robust to moderate asymmetry, but under strong asymmetry, tariffs on small trading partners can worsen the tariffing country's terms of trade and reduce welfare when foreign suppliers are sufficiently close substitutes. We use the framework to contrast exchange-rate responses during the 2025 tariff episode.

Keywords: tariffs, exchange rates, trade diversion, terms of trade, multilateral Marshall–Lerner condition.
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1 Introduction

What happens to exchange rates when a country imposes a tariff? In the standard two-country model, a tariff shifts expenditure away from imports and improves the trade balance at the prevailing exchange rate. Under the Marshall–Lerner condition, external balance is restored through an appreciation of the tariffing country's currency (Dornbusch, 1974; Krugman, 1982). Because there is only one partner, the bilateral exchange rate is also the effective exchange rate. In the long-run real environment considered here, it also determines the terms of trade. Bilateral appreciation, effective appreciation, and a terms-of-trade improvement therefore coincide.

These concepts separate once trade is multilateral. A tariff on one partner changes not only the relative price of foreign and domestic goods, but also relative prices across foreign suppliers. It can therefore affect the exchange rate against the target, the rates against untariffed bystanders, and the effective exchange rate differently. In any differentiable multi-country trade environment, we show that the first-order response can be represented as the adjustment of exchange rates to a vector of tariff-induced trade-balance disturbances. We define a multilateral Marshall–Lerner condition, implied by gross substitutability of trade balances in currency values, that ensures well-behaved adjustment but does not determine the sign of an individual bilateral response.

We first characterize the symmetric benchmark. When the tariffing country's partners are symmetric, we show that any tariff change separates into average protection and relative treatment. Average protection determines the nominal effective exchange rate, while relative treatment determines the dispersion of bilateral rates around it. Under a unilateral tariff, the two components reinforce one another against the target, but work in opposite directions against untariffed bystanders.

The bystander response depends on substitution toward domestic goods relative to substitution across foreign suppliers. Expenditure switching toward domestic goods supports the conventional appreciation. Diversion toward untariffed suppliers instead strengthens their currencies relative to the tariffing country. A countervailing force operates through the target. As the target loses access to the tariffing country's market, its currency depreciates, making its goods cheaper for third countries and placing downward pressure on bystander currencies. We establish that the tariffing country depreciates against a bystander only when substitution across foreign suppliers is sufficiently strong relative to domestic diversion. The reversal requires foreign suppliers to be closer substitutes for one another than foreign goods are for domestic goods and cannot arise, under standard home-bias conditions, in a specification with a single elasticity across all tradable varieties. Empirical estimates

that distinguish cross-foreign from home-foreign substitution support treating these margins separately (Broda and Weinstein, 2006; Feenstra et al., 2018; Li et al., 2022).

At the symmetric free-trade benchmark, we prove that a unilateral tariff appreciates the tariffing country's currency against the target and appreciates its nominal effective exchange rate under the multilateral Marshall–Lerner condition. Yet the tariff can depreciate the currency against every untariffed bystander when cross-origin substitution is sufficiently strong. We also establish an exact first-order reduction to a three-region economy consisting of the tariffing country, the target, and a composite bystander. The reduction shows that the number of countries matters not simply because the bystander bloc is larger, but because more of the bloc's trade is internal and highly substitutable. This result qualifies the competing-exporter logic in the preferential-trade literature, under which protection against one exporter improves the terms of trade of an excluded exporter (Viner, 1950; Mundell, 1964). When the importing country also produces a competing good, that third-country effect becomes conditional on foreign substitution being sufficiently strong. Despite a possible depreciation against bystanders, we prove that a unilateral tariff at the symmetric benchmark improves the tariffing country's aggregate terms of trade and raises welfare to first order.

Retaliation changes these conclusions. Under symmetry, we prove that equal bilateral tariffs leave the exchange rate between the two belligerents unchanged to first order, while adjustment occurs against third countries. The degree of foreign substitution required for depreciation is then substantially lower than under a unilateral tariff. When substitution across foreign suppliers exceeds the domestic-diversion margin, both belligerents depreciate against bystanders and in effective terms. In a multilateral trade war, the same condition leads the tariffing country to depreciate against every partner. The real trade channel can therefore generate effective appreciation under unilateral protection, bilateral stability under retaliation, and broad depreciation during a trade war without nominal or financial frictions.

The 2025 tariff episode provides the immediate motivation for revisiting the conventional benchmark. The dollar weakened against several major currencies following the tariff announcements while moving comparatively little against the renminbi. Several early analyses took the conventional appreciation result as their starting point and interpreted dollar weakness primarily through risk premia, convenience yields, valuation effects, or tariff uncertainty (Itskhoki and Mukhin, 2025; Kalemli-Özcan et al., 2026). These mechanisms may have been important and are complementary to our analysis. Our point is narrower: the two-country result is not sufficient to conclude that a multilateral depreciation must originate outside the trade channel.

The empirical evidence also highlights the role of policy configuration. Ostry et al.

(2025) document that the dollar tends to appreciate following unilateral US tariffs, depreciate when partners retaliate, and remain broadly stable against the renminbi during US–China tit-for-tat episodes. They interpret this pattern through dominant-currency pricing. We establish the same qualitative configuration dependence through real trade diversion at the symmetric benchmark. A unilateral tariff appreciates the effective exchange rate, while bilateral retaliation can leave the belligerents' exchange rate unchanged and depreciate both currencies against bystanders and in effective terms. The two mechanisms can coexist.

We use the model to provide a stylized interpretation of the 2025 episode, not a quantitative explanation or empirical test. The early phase combined discriminatory US tariffs on China with tariffs and retaliation in North America. The relative stability of the dollar against China and its weakness against important bystanders are consistent with the trade channel when substitution across foreign suppliers is strong, although the mixed policy configuration prevents a clean interpretation. The subsequent US–China escalation more closely approximated bilateral retaliation, for which our symmetric results predict a stable bilateral rate together with depreciation against bystanders and in effective terms. Later, US tariffs became more nearly uniform across partners. Our general uniform-tariff result implies that the increasingly uniform component should have supported effective appreciation, while remaining tariff differentials and retaliation worked in the opposite direction. The absence of clear appreciation in that phase leaves a role for forces outside the model. The episode must therefore be assessed phase by phase rather than against a single two-country prediction.

Several aggregate relationships continue to overlap under symmetry. Although bilateral rates can move differently from the effective rate, the trade weights used in the nominal effective exchange rate coincide at the symmetric benchmark with the import weights relevant for the aggregate terms of trade. Effective appreciation, a terms-of-trade improvement, and the first-order welfare gain therefore remain aligned. The exact composite-bystander representation also continues to apply. Local comparative statics and numerical exercises indicate that the main symmetric sign results are robust to moderate asymmetry, although the relevant thresholds vary with country size, trade weights, and home bias.

Strong asymmetry can overturn these aggregate conclusions. Away from symmetry, bilateral rates, the trade-weighted effective exchange rate, and the import-weighted aggregate terms of trade use different weights and need not move together. The bystander bloc can no longer generally be represented as a single country, and the relationship between foreign substitution and exchange-rate adjustment need not be monotonic. We show that when the target is small or carries little weight in the tariffing country's imports, and foreign suppliers are sufficiently close substitutes, depreciations against more heavily weighted bystanders can outweigh the appreciation against the target. A unilateral tariff can then worsen the aggre-

gate terms of trade and reduce welfare. Under stronger asymmetry, when bystanders also account for most of the tariffing country's trade, we show that even the nominal effective appreciation obtained under symmetry can reverse.

These reversals apply to discriminatory tariffs, not to uniform protection. Under general conditions allowing country sizes, trade shares, home bias, and preference parameters to differ, we prove that a small uniform tariff imposed from a balanced free-trade baseline appreciates the tariffing country's currency weakly against every partner under the multilateral Marshall–Lerner condition. It therefore improves the aggregate terms of trade and raises welfare to first order. This result recovers the classical terms-of-trade argument for uniform protection while showing why its partner-specific counterpart can fail. Uniform protection eliminates relative treatment across suppliers and switches off the cross-origin diversion mechanism that drives the reversals.

1.1 Related literature

A rapidly growing literature studies the exchange-rate effects of the 2025 tariffs through financial channels, retaliation and rich quantitative models; what it lacks, and what we provide, is a characterization of what the real trade channel *itself* implies for bilateral and effective exchange rates once trade is multilateral.

Tariffs and exchange rates. The classical two-country analysis of tariffs under flexible exchange rates (Lerner, 1936; Metzler, 1949; Mundell, 1961; Dornbusch, 1974; Eichengreen, 1981; Krugman, 1982) delivers the appreciation result; Dornbusch (1974) is particularly close to us in giving non-traded goods a role in the incidence of a tariff, which enters our domestic diversion margin through the rebate. Costinot and Werning (2019) give the modern statement of Lerner symmetry; since our results concern the real allocation, they are unaffected by Lerner-type reformulations of the policy. A large open-macroeconomy literature studies tariffs in the short run with nominal rigidities, where monetary policy, invoicing and the fiscal use of revenue shape the response (Lindé and Pescatori, 2019; Barattieri et al., 2021; Erceg et al., 2023; Jeanne, 2021; Bergin and Corsetti, 2023; Auray et al., 2025; Bianchi and Coulibaly, 2025; Werning et al., 2025). We complement that work with the long-run real benchmark in a multilateral setting, the position toward which those short-run dynamics converge once prices have adjusted.

The 2025 dollar. The financial and uncertainty explanations of the 2025 dollar (Kalemli-Özcan et al., 2026; Acharya and Laarits, 2026; Itskhoki and Mukhin, 2025; Cardani et al., 2025; Jiang et al., 2025) are complements to our analysis, not competitors; Kalemli-Özcan

et al. (2026), for instance, work in a two-country model, where tariff levels appreciate the currency and tariff volatility depreciates it. Itskhoki and Mukhin (2025) are the closest theory paper on the dollar: in their static benchmark a tariff appreciates the currency, and with dollar-denominated US liabilities that appreciation carries a valuation cost that sharply lowers the optimal macro tariff; a tariff war can then cost the United States its exorbitant privilege. Their mechanism is valuation and convenience yield; ours is real trade diversion. The paper closest to ours empirically is Ostry et al. (2025). Building a database of tariff announcements and retaliations for 2018–20 and 2025, they show that the dollar appreciates on unilateral US tariffs and depreciates when partners retaliate, while in US–China tit-for-tat exchanges it is roughly flat bilaterally. They interpret the pattern through dominant-currency pricing (Gopinath et al., 2020). Our bilateral-retaliation result predicts exactly the same pattern, with $d\nu_B = 0$ against the retaliating partner and depreciation against bystanders whenever $\rho > \rho_D$, without any nominal friction. The two explanations can coexist. Ours shows that the configuration-dependence they document is what the real trade channel requires in the long run, so it need not be read as evidence of a nominal or financial friction, and it identifies the substitution conditions under which that reading is warranted.

Third-country effects. Because the exchange rate *is* the terms of trade in our model class, the preferential-trade-agreement literature on third-country terms-of-trade effects applies directly (Viner, 1950; Mundell, 1964; Kemp and Wan, 1976; Krugman, 1991; Bond and Syropoulos, 1996; Bagwell and Staiger, 1999, 2002; Yi, 2000; Chang and Winters, 2002; Ornelas, 2005). Our contribution to it is narrow: the competing-exporter result, in which the excluded exporter's terms of trade always improve, becomes conditional ($\rho > N\rho_D$) once the importer also produces a competing good. Quantitative and empirical work on the third-country effects of trade wars (Darracq Pariès et al., 2026; Ignatenko et al., 2025; Fajgelbaum et al., 2024; Freund et al., 2024; Alfaro and Chor, 2023; Fajgelbaum and Khandelwal, 2026) quantifies these effects in rich environments; Darracq Pariès et al. (2026) conclude that tariff incidence is intrinsically multilateral and cannot be inferred from two-country analysis. These papers quantify; we characterize signs. Our uniform-tariff result also connects to optimal-tariff theory: a country with market power gains from a small tariff (Johnson, 1953; Broda et al., 2008; Costinot et al., 2015), and we recover this for a uniform tariff at any asymmetric baseline, whereas the partner-by-partner version can fail.

Elasticities. Our thresholds turn on the gap between the home–import (macro) and cross-origin (micro) elasticities. The evidence that cross-foreign elasticities exceed home–foreign

ones (Broda and Weinstein, 2006; Feenstra et al., 2018; Li et al., 2022), and that long-run trade elasticities are larger than short-run ones (Boehm et al., 2023), is exactly the distinction our results require.

Multi-market stability. The MML condition descends from the Hicks–Metzler–Mosak conditions for the stability of multiple markets and from the gross-substitutes stability results (Hicks, 1939; Mosak, 1944; Metzler, 1945; Arrow and Hurwicz, 1958; Arrow et al., 1959). We apply that lineage to the tariff–exchange-rate mapping rather than inventing the condition; the IMF's multilateral exchange rate model (Artus and Rhomberg, 1973) is an early institutional antecedent of multilateral exchange-rate modelling of this kind.

Roadmap. This paper is organized as follows. Section 2 develops the general disturbance-adjustment framework and the multilateral Marshall–Lerner condition, and derives the symmetric decomposition into average protection and relative treatment. Section 3 gives the decomposition economic content in a nested-CES model, establishes the unilateral-tariff results, and derives the exact three-region representation. Section 4 studies bilateral retaliation, multilateral trade wars, and preferential tariffs. Section 5 introduces asymmetric trade relationships, distinguishes the effective exchange rate from the welfare-relevant aggregate terms of trade, establishes the general uniform-tariff result, and applies the framework to the 2025 episode. Section 6 concludes. The appendices contain the derivations, proofs, and robustness results.

2 Tariffs and Exchange-Rate Adjustment: A General Framework

2.1 The bilateral case

Consider two countries, A and B. Let e_{AB} denote units of A's currency per unit of B's, so that a rise is a depreciation of A, and write $\nu \equiv \log e_{AB}$. Country A levies an ad valorem tariff τ on imports from B. Equilibrium with balanced trade is $\mathrm{TB}_A(\nu, \tau) = 0$. Linearizing,

$$d\,\mathrm{TB}_A = \mathrm{TB}_{A\nu}\,d\nu + \mathrm{TB}_{A\tau}\,d\tau = 0 \quad \Longrightarrow \quad d\nu^* = -\frac{\mathrm{TB}_{A\tau}}{\mathrm{TB}_{A\nu}}\,d\tau, \tag{2.1}$$

where $\mathrm{TB}_{A\nu} \equiv \partial \mathrm{TB}_A/\partial\nu$ and $\mathrm{TB}_{A\tau} \equiv \partial \mathrm{TB}_A/\partial\tau$ at fixed ν. A tariff raises the domestic price of imports and, at unchanged exchange rates, improves A's trade balance ($\mathrm{TB}_{A\tau} > 0$). The response is then an appreciation if and only if $\mathrm{TB}_{A\nu} > 0$, that is, if and only if a depreciation

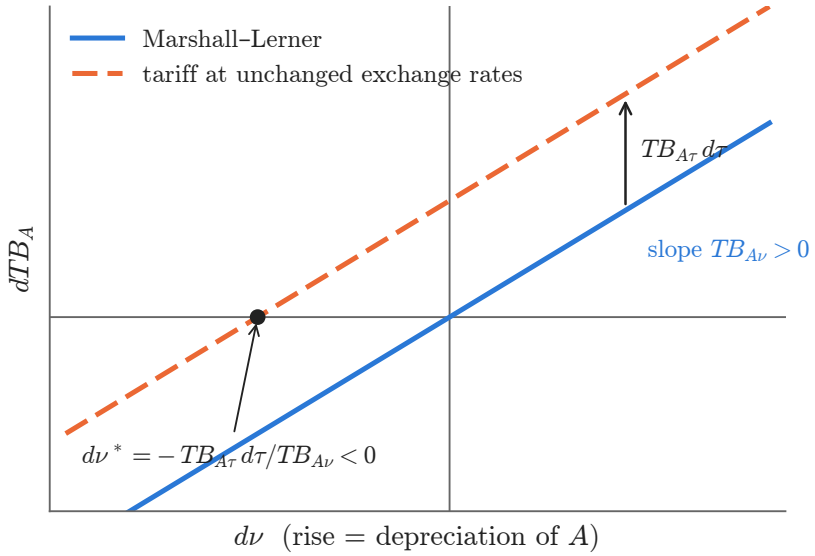


Figure 1: Bilateral tariff and exchange-rate adjustment. The solid line is A's trade balance as a function of the exchange rate at the initial tariff, with slope $\mathrm{TB}_{A\nu} > 0$ under Marshall–Lerner. The tariff shifts it up by $\mathrm{TB}_{A\tau}\, d\tau$ at unchanged exchange rates (dashed line), and balance is restored at $d\nu^* = -\mathrm{TB}_{A\tau}\, d\tau / \mathrm{TB}_{A\nu} < 0$, an appreciation of A. Schematic.

improves the trade balance. At a balanced baseline $\mathrm{TB}_{A\nu} = X(\varepsilon_X + \varepsilon_M - 1)$, where X is the value of exports and $\varepsilon_X, \varepsilon_M$ are the export and import demand elasticities, so this is the Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$. Figure 1 illustrates.

Remark 2.1 (Robustness of the two-country result). The sign $d\nu^*/d\tau < 0$ requires only that (i) import expenditure falls with its consumer price and the tariff does not reduce export revenue by more at fixed ν, and (ii) Marshall–Lerner holds. These conditions are satisfied under CES, nested-CES and quasi-linear preferences, under diminishing returns or sector-specific factors with upward-sloping export supply, and under any standard treatment of tariff revenue (Online Section D.1). The correct reading is not that appreciation holds in every model, but that **the two-country result holds wherever Marshall–Lerner holds**. The rest of the paper shows that with three or more countries this is no longer enough.

2.2 The disturbance–adjustment representation

Now consider N countries. Take A's currency as numeraire. The $N-1$ bilateral rates e_{Aj}, $j \neq A$, span all exchange-rate changes, and Walras' law leaves $N-1$ independent trade-balance conditions:

$$\mathrm{TB}_i\Big(\{e_{Aj}\}_{j\neq A}, \{\tau_{jk}\}_{j\neq k}\Big) = 0, \qquad i \neq A.$$

Stack $\nu = (\nu_j)_{j \neq A}$ with $\nu_j = \log e_{Aj}$, and the bilateral tariff changes as $d\tau = (d\tau_{jk})$. Linearizing,

$$J\, d\nu + G\, d\tau = 0, \qquad J_{il} \equiv \frac{\partial \mathrm{TB}_i}{\partial \nu_l}, \qquad G_{i,(jk)} \equiv \left.\frac{\partial \mathrm{TB}_i}{\partial \tau_{jk}}\right|_{\nu}. \tag{2.2}$$

Write a tariff change as a direction a ($\|a\| = 1$) times a scale ds, and let $F \equiv Ga$. Then

$$\boxed{\frac{d\nu}{ds} = (-J)^{-1} F.} \tag{2.3}$$

This separates two conceptually distinct objects. F is the **tariff incidence**: the vector of trade-balance disturbances at unchanged exchange rates, including expenditure switching, tariff revenue and income effects. $(-J)^{-1}$ is the **exchange-rate adjustment**: the movements needed to offset any given vector of disturbances. In the bilateral case both are scalars, and Eq. (2.3) reduces to Eq. (2.1). With $N \geq 3$, F can have entries of different signs, because a tariff on one partner may *improve* another partner's trade balance.

2.3 The multilateral Marshall–Lerner condition

Definition 2.2 (Multilateral Marshall–Lerner condition, MML)**.** The adjustment mechanism satisfies MML if $J_{il} \geq 0$ for $l \neq i$ and $(-J)^{-1}$ exists with $(-J)^{-1} \geq 0$ elementwise. Equivalently, $-J$ is a nonsingular M-matrix.

Under MML, writing $\vartheta_{ij} \equiv [(-J)^{-1}]_{ij} \geq 0$,

$$\frac{d\nu_i}{ds} = \sum_j \vartheta_{ij} F_j. \tag{2.4}$$

A positive disturbance in any country contributes weakly positively to every bilateral response. So **a uniformly signed F cannot generate responses of the opposite sign**, which is the multilateral version of the two-country logic. When F has mixed signs, the response of any bilateral rate depends on the whole vector of disturbances, not on its own entry.

A more primitive sufficient condition uses the full $N \times N$ Jacobian $\tilde{J}_{il} = \partial \mathrm{TB}_i / \partial \nu_l$ in terms of currency values, where now ν_l is the log value of currency l in a common unit of account. Homogeneity of degree one gives $\sum_l \tilde{J}_{il} = \mathrm{TB}_i = 0$ at a balanced baseline, and Walras' law gives $\sum_i \tilde{J}_{il} = 0$.

Assumption 2.1 (Gross substitutability of trade balances)**.** $\tilde{J}_{il} \geq 0$ for all $l \neq i$: an appreciation of currency l weakly improves every other country's trade balance.

Lemma 2.3. *If each trade balance is homogeneous of degree one in currency values, Assumption 2.1 holds, J is irreducible, and $\tilde{J}_{iA} > 0$ for some i, then MML holds.*

This is the trade-balance analogue of the gross-substitutes stability results of Hicks (1939), Mosak (1944), Metzler (1945) and Arrow et al. (1959). We use it as the multilateral counterpart of the Marshall–Lerner condition. The point to stress is that **MML does not pin down the sign of any individual exchange-rate response.** Every bilateral Marshall–Lerner condition can hold and the multilateral response can still go either way. What decides it is the structure of F.

2.4 Symmetric partners

The representation Eq. (2.3) holds at any balanced baseline. When the tariffing country's partners are symmetric, it can be solved in closed form in terms of four eigenvalues.

Restrict attention to tariff changes imposed by A, $t = (t_j)_{j \neq A}$ with $t_j \equiv d\tau_{Aj}$, and let G now denote the $(N-1) \times (N-1)$ restriction of the tariff-incidence matrix. Suppose A's partners are **symmetric**: $PJP' = J$ and $PGP' = G$ for every permutation P of the partners, as at free trade or under a common external tariff. Decompose

$$t = \bar{t}\,\mathbf{1} + \tilde{t}, \qquad \bar{t} = \tfrac{1}{N-1}\mathbf{1}'t, \qquad \mathbf{1}'\tilde{t} = 0. \tag{2.5}$$

We call $\bar{t}$ **average protection** and $\tilde{t}$ **relative treatment**. Any permutation-invariant matrix has the form $aI + b\mathbf{1}\mathbf{1}'$, so $\mathbf{1}$ and every zero-sum vector are eigenvectors of both G and $-J$:

$$G\mathbf{1} = g_S\mathbf{1}, \quad Gv = g_D v; \qquad (-J)\mathbf{1} = j_S\mathbf{1}, \quad (-J)v = j_D v \qquad (\mathbf{1}'v = 0).$$

Proposition 2.4 (Average protection and relative treatment)**.** *Let partners be symmetric, $N \geq 3$, and $j_S, j_D \neq 0$. For any tariff vector t,*

$$d\nu = \lambda_A\,\bar{t}\,\mathbf{1} + \lambda_R\,\tilde{t}, \qquad \lambda_A = g_S/j_S, \quad \lambda_R = g_D/j_D. \tag{2.6}$$

Hence, with $d\nu_A^E \equiv \frac{1}{N-1}\sum_{j \neq A} d\nu_j$ (the equally weighted effective rate),

$$d\nu_A^E = \lambda_A \bar{t}, \qquad d\nu_j - d\nu_A^E = \lambda_R \tilde{t}_j, \qquad d\log e_{jk} = \lambda_R(t_k - t_j). \tag{2.7}$$

Moreover, MML is equivalent to $j_S > 0$ and $j_D \geq j_S$.

Average protection alone moves the effective rate; relative treatment alone

moves bilateral rates around it and the cross rates among partners. The geometry is simple. The response vector lives in the plane spanned by the common direction **1** and the zero-sum directions, and each component is scaled by its own coefficient (Fig. 2).

3 The Nested-CES Model

We now specify an economy in which these eigenvalues, and hence every response in Section 2.4, can be computed from primitives.

3.1 Production, preferences and equilibrium

Each country $i \in \{1, \dots, N\}$ is endowed with L_i units of labor, its size, which is mobile across sectors. It produces a country-specific tradable and a non-tradable with labor only, $Y_{Ti} = A_{Ti}L_{Ti}$ and $Y_{Ni} = A_{Ni}L_{Ni}$, with $L_{Ti}+L_{Ni} = L_i$. Normalizing each tradable's producer price to one in its own currency, $w_i = A_{Ti}$ and $P_{Ni} = A_{Ti}/A_{Ni}$. Relative international producer prices therefore move only through exchange rates, and **the bilateral exchange rate equals the bilateral terms of trade and the relative wage in a common currency.**

Preferences are

$$C_i = C_{Ti}^{\alpha_T} C_{Ni}^{1-\alpha_T}, \quad C_{Ti} = \left[\alpha_D^{1/\eta} C_{Tii}^{\frac{\eta-1}{\eta}} + (1-\alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}}, \quad M_i = \left[\textstyle\sum_{j\neq i} \beta_{ij}^{1/\rho} C_{Tji}^{\frac{\rho-1}{\rho}}\right]^{\frac{\rho}{\rho-1}}. \tag{3.1}$$

Here η governs substitution between domestic tradables and the import composite, and ρ substitution across foreign origins; the latter is inactive when $N = 2$. Country i levies τ_{ij} on imports from j, so the consumer price is $p_{ij} = (1+\tau_{ij})e_{ij}$, and the import and tradable price indices are $P_{Mi} = [\sum_{j\neq i} \beta_{ij} p_{ij}^{1-\rho}]^{1/(1-\rho)}$ and $P_{Ti} = [\alpha_D + (1-\alpha_D)P_{Mi}^{1-\eta}]^{1/(1-\eta)}$. Tariff revenue is rebated lump-sum, so income is

$$I_i = \frac{w_i L_i}{1 - \sum_{j\neq i} \frac{\tau_{ij}}{1+\tau_{ij}} s_{ij}}, \tag{3.2}$$

with s_{ij} the realized tariff-inclusive expenditure share of income on j's good. Trade balances are measured at border prices,

$$\mathrm{TB}_i = \sum_{k\neq i} f_{ki} - \sum_{j\neq i} f_{ij}, \qquad f_{ij} = \frac{s_{ij}\tilde{I}_i}{1+\tau_{ij}}, \tag{3.3}$$

with $\tilde{I}_i = e_{Ai} I_i$ income in A's currency. The NEER uses baseline bilateral trade weights,

$E_i = \prod_{j\neq i} e_{ij}^{w_{ij}}$ with $w_{ij} = (f_{ij} + f_{ji})/\sum_k (f_{ik} + f_{ki})$; throughout, "effective exchange rate" refers to this trade-weighted NEER. The real exchange rate is $q_{ij} = e_{ij}P_j/P_i$, with P_i the consumer price index.

The **symmetric benchmark** is free trade with $\beta_{ij} = 1/(N-1)$, common parameters, and equal endowments and productivities. It retains home bias α_D. Write $m \equiv \alpha_T(1-\alpha_D)$ for the import share of income.

3.2 The two-country case

With one foreign supplier, linearizing gives $d\mathrm{TB}_A = mM_2\, d\nu_B + m\rho_D\, d\tau$, so

$$\left.\frac{d\log e_{AB}}{d\tau}\right|_{\tau=0} = -\frac{\rho_D}{M_2}, \qquad \rho_D \equiv \underbrace{\alpha_D\eta}_{\text{domestic substitution}} + \underbrace{(1-\alpha_D)(1-\alpha_T)}_{\text{rebate / non-tradables}}, \qquad M_2 \equiv 1 + 2\alpha_D(\eta-1) = \varepsilon_X + \varepsilon_M - 1. \tag{3.4}$$

ρ_D is the **domestic diversion margin**. It measures how strongly a tariff pulls expenditure toward the domestic economy, through substitution toward the domestic tradable and through the rebate's income effect on non-tradables. An exchange-rate movement generates no revenue, so it has no rebate term. This is why $\rho_D \neq M_2$.

3.3 The N-country case

We now evaluate the eigenvalues of Section 2.4 in this model.

Theorem 3.1 (Exchange-rate responses in the symmetric nested-CES model)**.** *At the symmetric benchmark define*

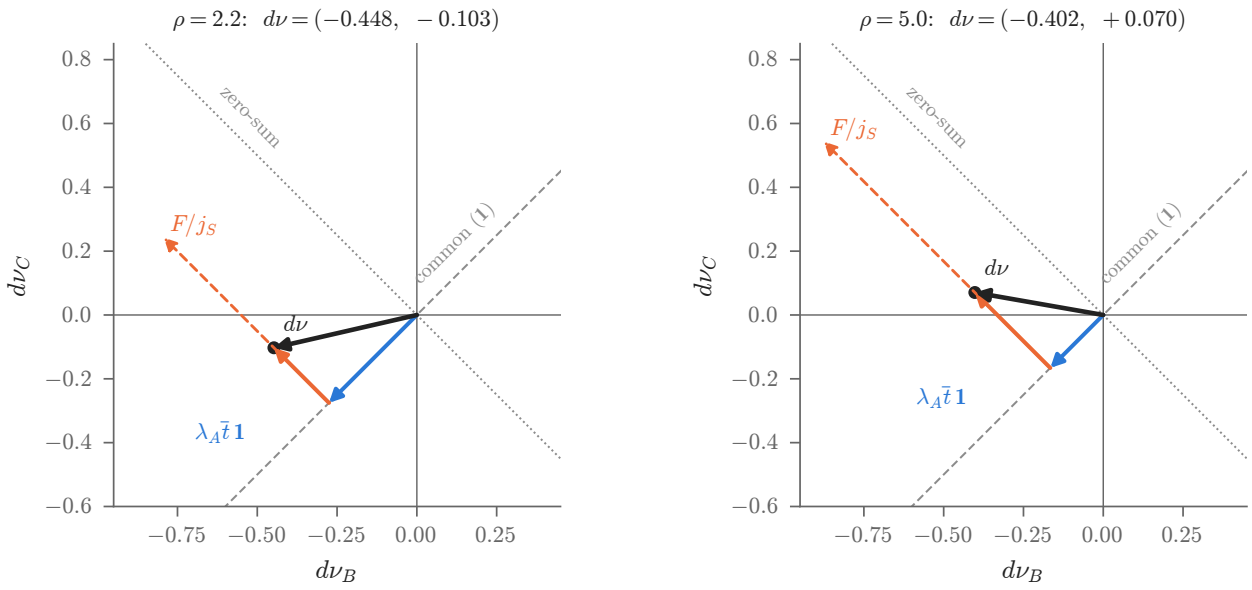
$$M_N \equiv \frac{1}{N-1}\Big[1 + \underbrace{N\alpha_D(\eta-1)}_{\textit{domestic–import adjustment}} + \underbrace{(N-2)\rho}_{\textit{foreign-origin adjustment}}\Big]. \tag{3.5}$$

The eigenvalues of Proposition 2.4 are (Lemma A.1 and Eq. (A.4))

$$g_S = -\frac{m\rho_D}{N-1}, \qquad g_D = -\frac{m\rho}{N-1}, \qquad j_S = \frac{mM_N}{N-1}, \qquad j_D = Nj_S. \tag{3.6}$$

MML holds if and only if $M_N > 0$, and $\eta \geq 1$ suffices. When it holds,

$$\lambda_A = -\frac{\rho_D}{M_N}, \qquad \lambda_R = -\frac{\rho}{NM_N}, \qquad d\nu_j = \lambda_A \bar{t} + \lambda_R \tilde{t}_j. \tag{3.7}$$



(a) $\rho = 2.2$: $\rho_D < \rho < 3\rho_D$ **(b)** $\rho = 5$: $\rho > 3\rho_D$

Figure 2: The geometry of average protection and relative treatment ($N = 3$). A unilateral tariff $t = (1, 0) = (\frac{1}{2}, \frac{1}{2}) + (\frac{1}{2}, -\frac{1}{2})$, with $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$ ($\rho_D = 1.17$, $3\rho_D = 3.51$). Blue: common component $\lambda_A \bar{t}\,\mathbf{1}$; solid orange: relative component $\lambda_R \tilde{t}$; black: equilibrium $d\nu$. The dashed orange arrow scales the relative component by N to show F/j_S. In (a), $\rho_D < \rho < 3\rho_D$: C's trade balance improves at unchanged exchange rates ($F_C > 0$), yet A still appreciates against C. In (b), $\rho > 3\rho_D$: A depreciates against C. In both, the NEER appreciates. Computed from Theorem 3.1.

M_N measures how sensitive A's trade balance with the rest of the world is to its effective exchange rate: $d\mathrm{TB}_A = mM_N\, d\nu_A^E$. When $N > 2$ it includes ρ, because foreign buyers can substitute between A's goods and other foreign goods. For $N = 2$ it reduces to M_2. Average protection works as in the two-country case. Relative treatment is zero-sum, so to first order it leaves revenue, income and the tradable/non-tradable split unchanged, and its disturbance is governed by ρ alone.

The factor $1/N$ in λ_R is central. A common exchange-rate move shifts only partners' trade with A. A relative move also shifts partners' trade with *each other*. At the symmetric benchmark A accounts for a share $1/N$ of the trade on which relative movements act, so $j_D = Nj_S$. **Relative exchange-rate adjustment is N times as effective as common adjustment**, and a given relative disturbance requires only $1/N$ as much exchange-rate movement.

3.4 A unilateral tariff

Let A raise its tariff on B alone, $t = d\tau\, e_B$. Then $\bar{t} = d\tau/(N-1)$, $\tilde{t}_B = \frac{N-2}{N-1} d\tau$ and $\tilde{t}_C = -\frac{d\tau}{N-1}$ for every bystander C.

Proposition 3.2 (Unilateral tariff)**.** *At the symmetric benchmark under MML,*

$$\frac{d\nu_A^E}{d\tau} = -\frac{\rho_D}{(N-1)M_N} < 0, \qquad \frac{d\nu_B}{d\tau} = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N} < 0, \qquad \frac{d\nu_C}{d\tau} = \frac{\rho - N\rho_D}{N(N-1)M_N}. \tag{3.8}$$

(i) A's NEER appreciates for every N and ρ.

(ii) A appreciates against the target.

(iii) A ***depreciates against each untariffed bystander if and only if*** $\rho > N\rho_D$***.***

(iv) The same threshold applies when A tariffs any $k \leq N-2$ partners. More generally, A depreciates against a partner j facing a below-average tariff increase if and only if $t_j/\bar{t} < 1 - N\rho_D/\rho$, which requires $\rho > N\rho_D$.

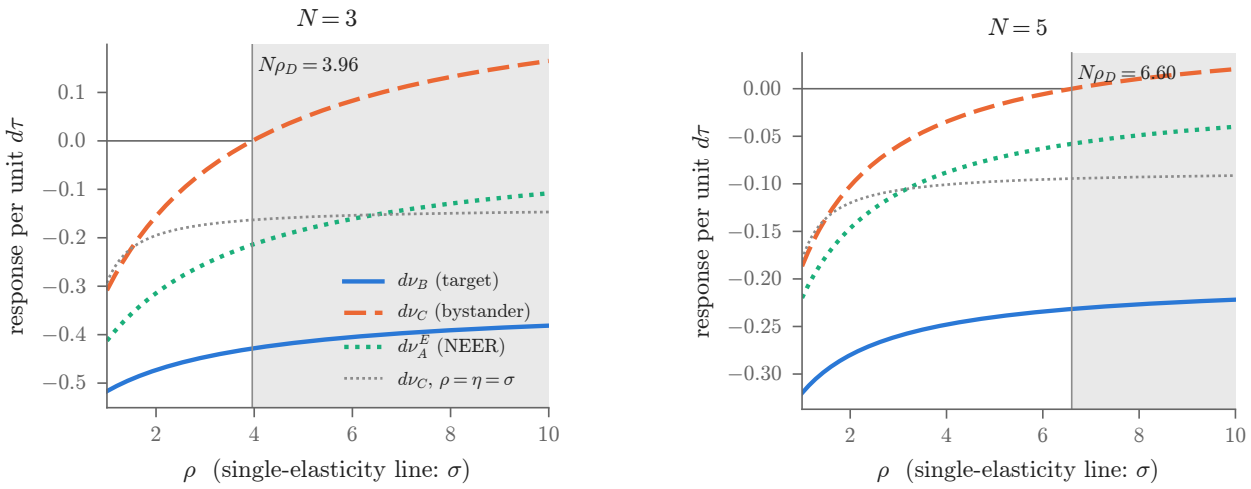
Against the target, average protection and relative treatment reinforce each other. Against a bystander they pull apart. Average protection shifts expenditure toward the domestic economy (appreciation), while the bystander's favorable relative treatment redirects expenditure toward it (depreciation).

Corollary 3.3 (Why a single elasticity cannot deliver reversal)**.** *With home bias $\alpha_D \geq 1/N$,*

$$N\rho_D - \eta = \eta(N\alpha_D - 1) + N(1-\alpha_D)(1-\alpha_T) \geq 0, \tag{3.9}$$

so $\rho > \eta$ is ***necessary*** *for a bystander depreciation. In any specification with a single elasticity across all tradable varieties ($\rho = \eta = \sigma$), A appreciates against every bystander at every σ. At $\alpha_D = 1/N$ the response is $-(1-\alpha_T)/(N\sigma)$, which approaches zero from below but never crosses it.*

This explains why the natural first extension of the two-country model, a third country under a single CES elasticity, has not surfaced the reversal. The reversal requires that cross-origin substitution exceed home–import substitution, an ordering supported by the macro–micro elasticity evidence (Feenstra et al., 2018; Li et al., 2022). At US-like parameters ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$, so $\rho_D = 1.32$), $N\rho_D = 3.96$ for $N = 3$.



(a) $N = 3$ **(b)** $N = 5$

Figure 3: Bilateral and effective responses to a unilateral tariff, per unit $d\tau$, US-like parameters ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$). $d\nu_B$ and $d\nu_A^E$ are negative for every ρ; $d\nu_C$ crosses zero at $N\rho_D$ (3.96 for $N = 3$, 6.60 for $N = 5$), and the shaded region is $\rho > N\rho_D$. The grey dotted line is the single-elasticity case $\rho = \eta = \sigma$, plotting $d\nu_C$ against σ; it never crosses zero (Corollary 3.3). Closed forms of Proposition 3.2, verified against the linear solver.

Disturbance versus equilibrium. At unchanged exchange rates the bystander's trade balance improves if and only if $\rho > \rho_D$:

$$F_C = \frac{m(\rho - \rho_D)}{(N-1)^2}. \tag{3.10}$$

The equilibrium bystander depreciation requires the stronger condition $\rho > N\rho_D$. The gap is the $1/N$ effectiveness of relative adjustment. For $\rho_D < \rho < N\rho_D$, C gains at the old exchange rates yet ends up with a weaker currency against A (Fig. 2a). Figure 3 plots the three responses against ρ.

3.5 The three-country representation

The N-country results can be read through a three-country economy, which we use as the workhorse for intuition. When the $N - 2$ bystanders are symmetric and the tariff shock preserves that symmetry, no relative-price movement within the bystander bloc is activated. The bloc then has a single active direction, and the tariff experiment reduces exactly to three regions: A, B and a composite bystander C. We now show that this composite is itself a nested-CES country, and we identify its parameters.

Proposition 3.4 (Exact three-country reduction)**.** *At the symmetric benchmark, consider any tariff change among A, B and the bystanders that is common across bystanders: tariffs*

by A or B on all bystanders, by all bystanders on A or B, or between A and B. To first order, the responses of ν_B, of every bystander rate ν_C and of A's NEER coincide with those of a three-country nested-CES economy $\{A, B, C\}$ in which:

- *C has income $y_C = N - 2$;*
- *A and B place import weight $\frac{N-2}{N-1}$ on C and $\frac{1}{N-1}$ on each other;*
- *C places import weight $\frac{1}{2}$ on each of A and B, with cross-origin elasticity ρ;*
- *C's home share of tradables is $h_C = \alpha_D + (1 - \alpha_D)\frac{N-3}{N-1}$, equivalently $(N-1)h_C = N - 3 + 2\alpha_D$;*
- *C's home–import elasticity is*

$$\boxed{\eta_C = \frac{(N-3)\rho + 2\alpha_D\,\eta}{N - 3 + 2\alpha_D}.} \tag{3.11}$$

Proof sketch. All bystander goods have the same price change, so CES aggregation of A's and B's demand over bystander goods is exact. Trade within the bloc cancels from the bloc's trade balance; what remains is the bloc's demand for A's and B's goods. Measure prices in bystander currency, so bloc goods have $dp = 0$, and write $S = dp_A + dp_B$ and $D = dp_A - dp_B$. A bystander's import price index moves by $S/(N-1)$, since A and B are 2 of its $N-1$ import origins. Hence

$$d\log s_{cA} = (1-\rho)\tfrac{D}{2} + \left[(1-\rho)\tfrac{N-3}{2(N-1)} + (1-\eta)\tfrac{\alpha_D}{N-1}\right]S.$$

In the composite the import index moves by $S/2$, so $d\log s_{CA} = (1-\rho)\frac{D}{2} + (1-\eta_C)h_C\frac{S}{2}$. The D terms coincide. Matching the S terms gives $(1-\eta_C)(N-1)h_C = (1-\rho)(N-3) + 2\alpha_D(1-\eta)$, which is Eq. (3.11). The full proof is in Section B. □

The reduction is first-order. At finite tariffs a CES with constant η_C does not replicate the bloc. The shocks it covers include the retaliation and trade-war experiments of Section 4.

Interpretation. The composite bystander differs from an ordinary large country in one respect. Most of each bystander's trade is with *other bystanders*, whose goods sit in the same ρ-nest as A's and B's. Once collapsed, this internal trade becomes the composite's "home" trade, and it is substitutable at a blend of ρ and η. When $\rho > \eta$, η_C rises toward ρ as N grows. The bystander bloc becomes inward-looking ($h_C \to 1$) and highly substitutable.

The reduced system. In the composite economy the 2×2 system for (ν_B, ν_C) is, with $c = mM_N/(N-1)$,

$$-J = c \begin{pmatrix} N-1 & -(N-2) \\ -(N-2) & 2(N-2) \end{pmatrix}, \qquad \det(-J) = c^2 N(N-2), \tag{3.12}$$

and

$$d\nu_C = \frac{F_B + (N-1)F_c}{cN}, \qquad d\nu_B = \frac{2F_B + (N-2)F_c}{cN}, \tag{3.13}$$

where F_B is the target's disturbance and F_c each bystander's ($F_C = (N-2)F_c$). Under a unilateral tariff, $F_B = -m[\rho_D + (N-2)\rho]/(N-1)^2 < 0$ and $F_c = m(\rho - \rho_D)/(N-1)^2$. Equation (3.13) recovers Eq. (3.8) and gives the bystander response a two-force reading:

- **diversion**, $(N-1)F_c$: expenditure diverted toward bystanders improves their position and strengthens their currencies;
- **redirection**, F_B: the target loses access to A's market and its currency must fall. This makes the target's goods cheaper for bystanders and drags bystander currencies down with it.

A depreciates against the bystander if and only if diversion outweighs redirection, $(N-1)(\rho - \rho_D) > \rho_D + (N-2)\rho$, which is $\rho > N\rho_D$. These are the "trade diversion" and "trade redirection" forces of earlier versions of this paper, now with exact weights.

Why N raises the threshold, and why size alone does not. One might expect that adding countries is equivalent to enlarging the bystander. It is not. Hold the composite's flows fixed at their N-country values but give it an ordinary home–import elasticity η instead of η_C. The bystander threshold then *falls* slightly with N, consistent with the general result (Section C.3) that a larger bystander lowers the threshold. The true threshold $N\rho_D$ rises steeply (Table 1 and Fig. 4). What drives the N-dependence is the bloc's internal, highly substitutable trade, which makes relative exchange-rate adjustment N times as effective as common adjustment and dilutes A's footprint to $1/N$.

3.6 Real exchange rates

Real rates add a price-index wedge: the tariff raises A's consumer prices directly. With $\bar{t}$ the average tariff and partners imposing no tariffs,

$$d\log q_{Aj} = \left(1 - \tfrac{mN}{N-1}\right) d\nu_j - m\bar{t}, \qquad dq_A^E = \left[\left(1 - \tfrac{mN}{N-1}\right)\lambda_A - m\right]\bar{t}. \tag{3.14}$$

Table 1: N versus bystander size

		Composite bystander threshold		
N	$N\rho_D$ (true)	with η_C Eq. (3.11)	with ordinary η	η_C at $\rho = 4$
3	3.96	3.96	3.96	1.50
4	5.28	5.28	3.82	2.46
5	6.60	6.60	3.76	2.89
8	10.56	10.56	3.71	3.39

Notes: US-like parameters ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$). Thresholds are the values of ρ at which the bystander response to a unilateral tariff changes sign in the composite economy of Proposition 3.4, computed with the linear system of Section A. The "ordinary η" column keeps the composite's flows but replaces η_C by η.

Proposition 3.5 (Real exchange rates)**.** *Consider a unilateral tariff at the symmetric benchmark.*

(i) If $m < 1/N$ and MML holds at the relevant ρ, the real bilateral threshold exceeds the nominal one:

$$\rho_q^*(N) = \frac{[(N-1) - mN]N\rho_D + mN[N\alpha_D(\eta - 1) + 1]}{(N-1)(1 - mN)} > N\rho_D.$$

(ii) If $m < (N-1)/N$, the REER appreciates.

At US-like parameters and $N = 3$, $\rho_q^* = 4.93$, against $3\rho_D = 3.96$.

4 Trade Wars and Preferential Tariffs

So far only A has changed its tariffs. This section lets partners respond, and then relates the bystander result to the literature on preferential tariffs.

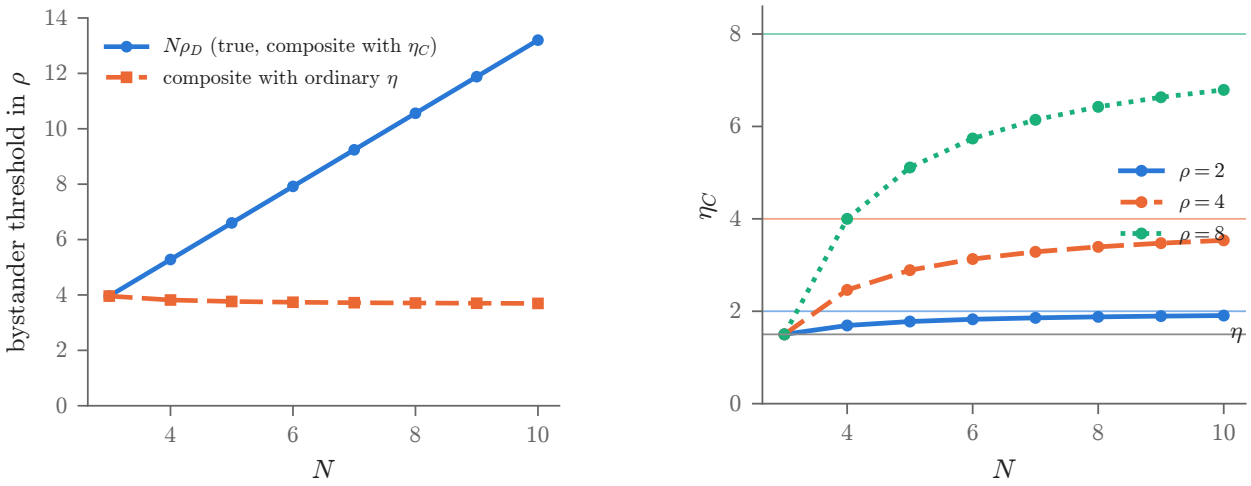
4.1 Trade wars

Since the analysis is local, simultaneous tariff changes by several countries add up.

Proposition 4.1 (Retaliation and trade wars)**.** *At the symmetric benchmark with $N \geq 3$ under MML:*

(i) ***Bilateral retaliation*** *(A tariffs B and B tariffs A by $d\tau$): $d\nu_B = 0$, and*

$$\frac{d\nu_C}{d\tau} = \frac{\rho - \rho_D}{(N-1)M_N}, \qquad \frac{d\nu_A^E}{d\tau} = \frac{N-2}{N-1} \cdot \frac{\rho - \rho_D}{(N-1)M_N}. \tag{4.1}$$



(a) Bystander threshold **(b)** Effective elasticity η_C

Figure 4: The composite bystander, US-like parameters. (a) The true threshold $N\rho_D$ for $N = 3, \ldots, 10$, which the composite with η_C reproduces exactly, and the threshold of the "ordinary-η" composite, which is nearly flat and slightly declining. (b) η_C from Eq. (3.11) against N for $\rho \in \{2, 4, 8\}$, converging toward ρ (thin horizontal lines); the grey line is $\eta = 1.5$.

Both belligerents depreciate against every bystander, and in effective terms, ***if and only if*** $\rho > \rho_D$*. Cross rates among bystanders are unchanged.*

(ii) ***Multilateral trade war*** *(A tariffs every partner, and every partner retaliates against A):* $d\nu_j = d\nu_A^E = \frac{(N-2)(\rho-\rho_D)}{(N-1)M_N}\, d\tau$ *for all* j*, so A depreciates against everyone if and only if* $\rho > \rho_D$*.*

B's retaliation adds a second relative-treatment effect in the bystander's favor, which offsets the $1/N$ dilution. The threshold drops from $N\rho_D$ to ρ_D, the point at which the bystander's trade balance improves at unchanged exchange rates. At US-like parameters $\rho_D = 1.32$, below essentially all cross-origin estimates. **Retaliation therefore makes multilateral depreciation of the tariffing country's currency the generic outcome, while leaving the belligerents' bilateral rate unchanged.** This is the pattern documented by Ostry et al. (2025), including the near-stability of the dollar against the renminbi in tit-for-tat episodes. Here it arises with flexible prices, balanced trade and no financial channel.

The two experiments differ in where adjustment falls. Under bilateral retaliation the belligerents' rate is fixed and adjustment runs against bystanders. In a multilateral war there are no bystanders, and all of A's rates move together (Fig. 5).

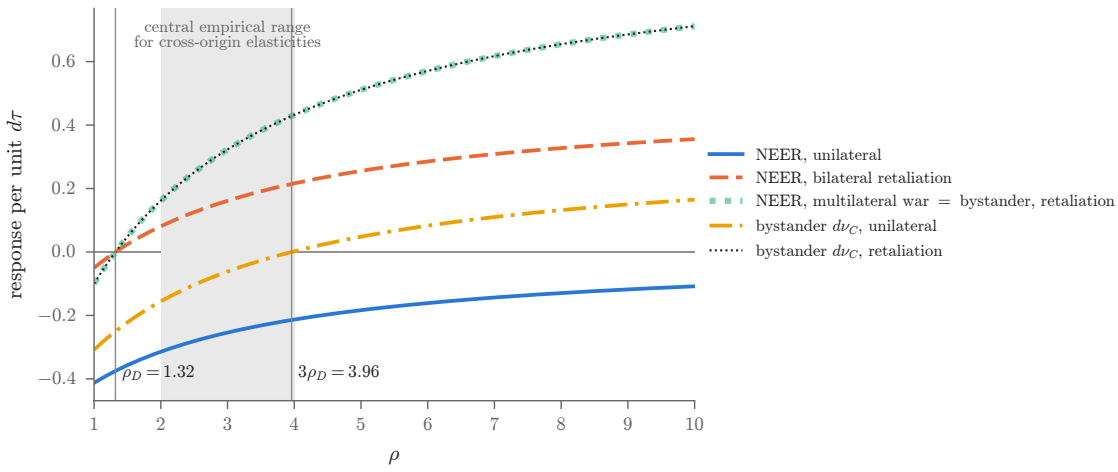


Figure 5: Configuration dependence, $N = 3$, US-like parameters. NEER responses $d\nu_A^E/d\tau$ under a unilateral tariff (always negative), bilateral retaliation and a multilateral war (both crossing zero at $\rho_D = 1.32$), and the bilateral response against a bystander under a unilateral tariff (crossing at $3\rho_D = 3.96$) and under retaliation (crossing at ρ_D). At $N = 3$ the multilateral-war response coincides with the bystander response under bilateral retaliation, so those two lines overlap. Shaded: the central empirical range for cross-origin elasticities, $\rho \in [2, 4]$.

4.2 Preferential tariffs and competing exporters

In our model class the bystander's exchange rate is its terms of trade: $\text{ToT}_{CA} = e_{AC}$. The response in Eq. (3.8) is therefore the third-country terms-of-trade effect of a discriminatory tariff, the central object of the preferential-trade-agreement literature since Viner (1950) and Mundell (1964). In the canonical competing-exporter model, an importer A that does not produce the good buys it from B and C under quasi-linear preferences. With q_j exporter j's price, E_j its export supply and M_A the importer's demand, a tariff on B then *always* improves C's terms of trade, since

$$\frac{dq_C}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} > 0,$$

and preferential liberalization toward B always hurts the excluded country. In our model the same effects hold **only if** $\rho > N\rho_D$. The difference is that A produces a competing tradable. Part of the demand diverted from B goes to A's own good, along with the rebate's income effect, so the third-country effect survives only when cross-origin substitution is strong relative to domestic diversion. The two environments are not formally nested. The competing-exporter model can be read as switching off ρ_D in order to isolate substitution across foreign suppliers. Section C.1 shows that at a positive common external tariff the threshold rises, and that a discrete preferential cut requires a higher ρ than the local threshold suggests.

5 Asymmetry

The symmetric case delivers clean geometry: the NEER and the bilateral rates separate exactly, and thresholds are linear in ρ. Asymmetry breaks the eigen-structure. This section does not aim at a general theory of asymmetric trade networks. Its purpose is to show which intuitive conclusions of the symmetric case are structural and which are artefacts of symmetry. The most consequential casualty is the result that a unilateral tariff always improves the tariffing country's aggregate terms of trade.

Two weighted averages of bilateral responses appear in this section and should be kept apart. The *effective exchange rate* is, as throughout the paper, the trade-weighted NEER, with weights w_{Aj} that combine A's imports from and exports to j. The *aggregate terms of trade* weight the same bilateral responses by A's import shares b_{Aj}. At the symmetric benchmark the two weightings coincide; away from it they do not, and it is the import-weighted aggregate terms of trade that is welfare-relevant.

5.1 The aggregate terms of trade and welfare

Let b_{Aj} be partner j's share in A's import bundle, and $R_j^B \equiv \partial \nu_j / \partial \tau_{AB}$. Define the import-weighted foreign-price response

$$R_{A|B} \equiv \sum_{j \neq A} b_{Aj} R_j^B. \tag{5.1}$$

Under the producer-price normalization, a rise in e_{Aj} raises the price of j's good relative to A's, so A's aggregate terms of trade improve if and only if $R_{A|B} < 0$.

Lemma 5.1 (First-order welfare)**.** *At a free-trade baseline, for any tariff vector, the first-order change in A's welfare, in units of income, is*

$$\frac{dU_A}{U_A' \, I_A} = -\, m_A \sum_{j \neq A} b_{Aj} \, d\nu_j, \tag{5.2}$$

where m_A is A's import share of income. A marginal tariff thus raises A's welfare if and only if it improves A's aggregate terms of trade.

Proof sketch. A's wage and non-tradable price are fixed in A's currency. By the envelope theorem, $\lambda \, dU_A = dT_A^{rev} - \sum_j C_{Aj} \, dp_{Aj}$. At free trade $dT_A^{rev} = \sum_j f_{Aj} \, d\tau_{Aj}$ and $dp_{Aj} = p_{Aj}(d\nu_j + d\tau_{Aj})$. The tariff terms cancel, leaving $-\sum_j f_{Aj} d\nu_j$. Foreign tariffs affect A only through $d\nu$. The full proof is in Section B. □

The welfare gradient. By Lemma 5.1, the first-order welfare effect is linear in the tariff vector. For tariffs imposed by A,

$$\frac{dU_A}{U'_A\, I_A} = \sum_{j \neq A} g_j \, d\tau_{Aj}, \qquad g_j \equiv -m_A \, R_{A|j}, \tag{5.3}$$

where $R_{A|j}$ is the import-weighted response Eq. (5.1) to a tariff on j alone. The component g_j is the welfare gain from a marginal tariff on partner j, and its sign is that of the aggregate terms-of-trade effect of that tariff. A uniform marginal tariff, $d\tau_{Aj} = dt$ for all j, is worth $\sum_j g_j \, dt$.

Proposition 5.2 (Uniform tariffs)**.** *Consider any balanced free-trade baseline at which MML holds; country sizes, origin shares, home biases and parameters may differ across countries. A uniform marginal tariff by A, $d\tau_{Aj} = dt > 0$ for all $j \neq A$, generates the disturbance*

$$F_j = -f_{Aj} \, \rho_D^A \, dt, \qquad \rho_D^A \equiv \eta_A h_A + (1 - h_A)(1 - \alpha_T^A) > 0, \tag{5.4}$$

where h_A is A's home share of tradable spending and η_A, α_T^A are A's own parameters. The tariff appreciates A's currency weakly against every partner, and strictly against every partner if A imports from all of them or J is irreducible. It therefore improves A's aggregate terms of trade and raises A's welfare to first order: $\sum_{j \neq A} g_j > 0$.

Proof sketch. At unchanged exchange rates A's tariffs move only the flows f_{Aj} (Eq. (A.1)). A uniform tariff raises every import price by the same amount, so relative import prices do not change and the cross-origin margin $(1 - \rho)$ drops out. Every import share then moves by $(1 - \eta_A)h_A \, dt$ and A's income by $m_A \, dt$, which gives Eq. (5.4). Since $F < 0$, Eq. (2.4) gives $d\nu \leq 0$, and Lemma 5.1 gives the welfare gain. The full proof is in Section B. □

At the symmetric benchmark $\rho_D^A = \rho_D$ and $f_{Aj} = m/(N-1)$, so Eq. (5.4) is the average-protection disturbance of Section 3.3. The proposition extends the conventional appreciation result from that benchmark to *any* balanced baseline. Every partner's trade balance is disturbed in the same direction, because a uniform tariff switches off cross-origin substitution, the force behind every reversal in this paper, and leaves only domestic diversion. By Eq. (2.4), a uniformly signed disturbance cannot produce responses of the opposite sign. The replication code confirms Eq. (5.4) to 10^{-12}, and the signs of F, $d\nu$ and the welfare change, on random asymmetric baselines with country-specific parameters and $N = 3, 4, 5$.

What the proposition does not say is that each component of the gradient is positive. A tariff on a single partner B does change relative import prices, and $R_{A|B}$ combines the

favorable move against B with the moves against everyone else, which can be adverse. Sections 5.2 and 5.3 show that $g_B < 0$ is possible. Since the components sum to a positive number, with $N = 3$ the condition $g_B < 0$ implies $g_C > 0$: whenever a tariff on the target lowers welfare, a tariff on the bystander raises it.

At the symmetric benchmark the components are equal, so each is positive. Import weights also equal trade weights there, so $R_{A|B} = d\nu_A^E/d\tau < 0$ (Proposition 3.2). **At the symmetric benchmark a unilateral tariff on one partner improves the tariffing country's aggregate terms of trade and raises its welfare to first order, even when it depreciates against every bystander.** Under bilateral retaliation the sign flips when $\rho > \rho_D$. For $N = 3$,

$$\frac{R_B^B + R_C^B}{2} = -\frac{\rho_D}{2M_3} :$$

the favorable move against the target always dominates the adverse move against the bystander when weights are equal.

Relation to optimal-tariff theory. A country with market power gains from a small tariff. This classic terms-of-trade argument is the starting point of Johnson (1953), who shows that a country with sufficient monopoly–monopsony power can remain better off than under free trade even after foreign retaliation. The argument has empirical support: Broda et al. (2008) find that countries not bound by WTO commitments set higher tariffs on goods whose foreign export supply is less elastic. Costinot et al. (2015) show, in a two-country Ricardian model with a continuum of goods, that optimal import tariffs are uniform across goods. Proposition 5.2 recovers the classic result for a small *uniform* tariff at any asymmetric baseline under MML, with the terms-of-trade gain delivered through the exchange rate. What fails is the partner-by-partner version. With many partners, a tariff on one of them also moves the terms of trade against the others. Nondiscrimination is what confines a country's tariffs to a single terms-of-trade channel (Bagwell and Staiger, 1999, 2002). Our results show a cost of discrimination from the imposing country's own point of view: even though A has market power against every partner, a tariff on a single partner can lower its welfare.

5.2 The aggregate terms-of-trade condition

Separate the target from the remaining suppliers, and let $\bar{R}_{-B}^B = \sum_{j \neq A,B} b_{Aj} R_j^B / (1 - b_{AB})$ be the conditional import-weighted bystander response. Then $R_{A|B} = b_{AB} R_B^B + (1 - b_{AB}) \bar{R}_{-B}^B$.

Definition 5.3 (Aggregate terms-of-trade condition)**.** Unilateral tariffs by A have the conventional aggregate terms-of-trade effect if, for every target B,

$$b_{AB}R_B^B + (1 - b_{AB})\bar{R}_{-B}^B < 0. \tag{5.5}$$

If $R_B^B < 0$ and $\bar{R}_{-B}^B > 0$, the condition fails if and only if

$$\frac{\bar{R}_{-B}^B}{-R_B^B} > \frac{b_{AB}}{1 - b_{AB}}. \tag{5.6}$$

The left side is equilibrium incidence: the adverse bystander move relative to the favorable move against the target. The right side is exposure: A's import dependence on the target relative to other suppliers. A bilateral bystander reversal is therefore **necessary but not sufficient** for an aggregate reversal. The adverse moves must also be large enough and carry enough weight in A's basket.

In terms of Eq. (5.3), the condition requires every component g_B of the welfare gradient to be positive, whereas Proposition 5.2 guarantees only their sum. The condition has a three-country form for any configuration of bystanders simply because $\bar{R}_{-B}^B$ is defined as their import-weighted average; what does not carry over under heterogeneous bystanders is the representation of the bloc as an ordinary country with primitive parameters (Section 5.4).

5.3 Asymmetric partners in the three-country model

In the three-country model the local system is $A(\rho)R = F(\rho)$ with $A \equiv -J$. At a given baseline every entry of A and F is affine in ρ, because ρ enters share responses only through $(1 - \rho)$. Since A is 2×2, its adjugate is affine too, and

$$R_{A|B}(\rho) = \frac{b' \operatorname{adj}(A(\rho))F(\rho)}{\det A(\rho)}, \qquad b' \operatorname{adj}(A(\rho))F(\rho) = q_2\rho^2 + q_1\rho + q_0 \equiv Q_{A|B}(\rho). \tag{5.7}$$

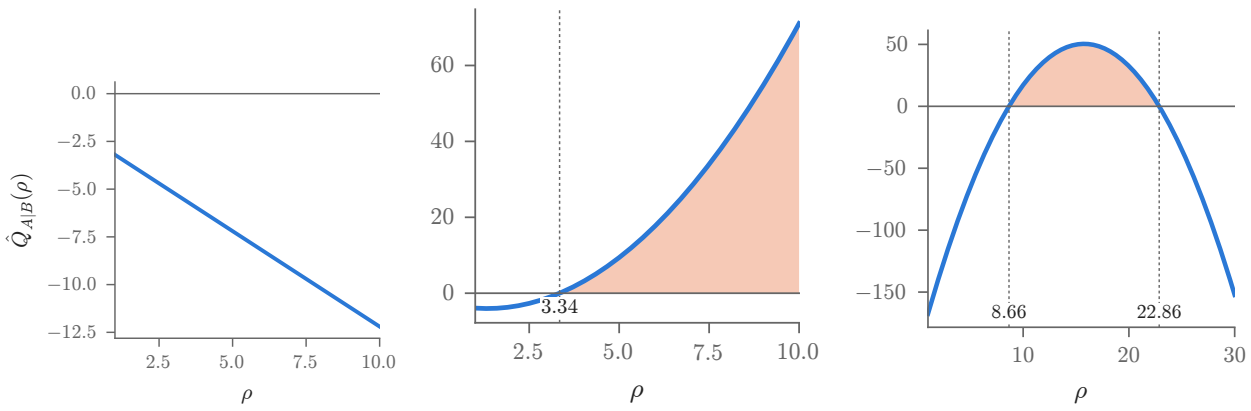
Under MML $\det A > 0$, so the aggregate condition is $Q_{A|B}(\rho) < 0$. The same structure governs the bilateral bystander response, whose numerator has leading coefficient

$$\alpha_T^2\, b_{AB}b_{AC}b_{CA}b_{CB}(1 - h_A)(1 - h_C)y_A y_C > 0$$

(with a common α_T), where h_i are home shares of tradable spending and y_i incomes.

Proposition 5.4 (Three qualitative cases)**.** *Holding the baseline fixed:*

(a) ***Symmetric:*** $q_2 = 0$ *and* $Q < 0$ *for all* ρ*. The condition always holds.*



(a) Symmetric **(b)** Small, low-weight target **(c)** Asymmetric network

Figure 6: Three relationships between ρ and the aggregate terms-of-trade condition. Each panel plots the normalized numerator $\hat{Q}_{A|B}(\rho)$ of Eq. (5.7) at a fixed baseline; the condition fails where $\hat{Q} > 0$ (shaded). (a) Symmetric benchmark: never crosses. (b) Example (b): single crossing at 3.34. (c) Example (c): crossings at 8.66 and 22.86. Signs verified against the linear solver.

(b) $q_2 > 0$, $q_0 < 0$*: exactly one positive root* ρ^**. The condition holds for* $\rho < \rho^*$ *and fails for* $\rho > \rho^*$*.*

(c) $q_2 < 0$, $q_1 > 0$, $q_0 < 0$, ***positive discriminant:*** *two positive roots. The condition fails only for intermediate* ρ*.*

Examples. All use $\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$ unless stated, with the free-trade baseline computed from primitives at $\bar{\rho} = 3\rho_D = 3.96$ and then held fixed; $\hat{Q}$ denotes Q normalized by $|q_2|$ (or by $|q_1|$ when $q_2 = 0$).

(a) Symmetric benchmark: $\hat{Q} = -(\rho + 2.2)$.

(b) Labor endowments $(L_A, L_B, L_C) = (1, 0.22, 1)$, $\beta_{AB} = 0.2$, $\beta_{AC} = 0.8$, other origin weights symmetric: $\hat{Q} = \rho^2 - 2.6695\rho - 2.2568$, so $\rho^* = 3.344$.

(c) A strongly asymmetric network, specified directly in baseline shares: $\alpha_T = 0.862$, $\eta = 3.007$, home shares $(0.201, 0.854, 0.360)$, origin shares $(b_{AB}, b_{AC}) = (0.525, 0.475)$, $(b_{BA}, b_{BC}) = (0.766, 0.234)$, $(b_{CA}, b_{CB}) = (0.407, 0.593)$, and incomes $(1, 5.126, 0.867)$, which balance trade at these shares. Then $\hat{Q} = -\rho^2 + 31.53\rho - 198.04$, so the condition fails for $8.66 < \rho < 22.86$.

Which asymmetries matter. Fix $L_A = L_C = 1$ and symmetric foreign-origin preferences for B and C. Vary the target's relative size L_B/L_C and its weight β_{AB} in A's imports, recomputing the free-trade equilibrium at $\bar{\rho} = 3\rho_D = 3.96$. In this family $q_0 < 0$ throughout

Table 2: Sign of q_2 under size and import-weight asymmetry

	β_{AB}						
L_B/L_C	0.10	0.20	0.35	0.50	0.65	0.80	0.90
0.1	+2.6	+3.1	+4.0	+5.4	+7.9	+15.8	+63.3
0.2	+2.7	+3.3	+4.4	+6.3	+10.2	+27.4	−
0.4	+3.0	+3.8	+5.5	+8.8	+18.9	−	−
0.7	+3.3	+4.6	+7.9	+18.2	−	−	−
1	+3.8	+5.8	+13.7	0	−	−	−
1.5	+4.8	+10.6	−	−	−	−	−
2	+6.8	+>100	−	−	−	−	−
3	−	−	−	−	−	−	−
5	−	−	−	−	−	−	−

Notes: $N = 3$, US-like parameters, $L_A = L_C = 1$, $\beta_{BA} = \beta_{CA} = 0.5$. The free-trade baseline is computed from primitives at $\bar{\rho} = 3.96$ and held fixed. "+" means $q_2 > 0$: the aggregate terms-of-trade condition eventually fails as ρ rises; the small number is the root ρ^* beyond which it fails. "−" means $q_2 < 0$, so the condition holds for all ρ in this family. "0" is the symmetric benchmark. $q_0 < 0$ and MML hold in every cell.

and the two-root case never appears. The sign of q_2 therefore classifies whether a high enough ρ eventually overturns the condition (Table 2). Positive entries fill the upper-left region, where the target is small, has a low import weight, or both, with an approximate trade-off between size and weight.

Takeaway. The aggregate terms of trade, and hence the first-order welfare gain from a tariff, are most likely to *deteriorate* when three forces align:

1. the target is small relative to the untariffed suppliers;
2. the target has a small weight in the tariffing country's imports;
3. foreign suppliers are close substitutes (high ρ).

In that region the bystander depreciation is both large and heavily weighted, while the gain against the target is small and lightly weighted. **A country can lower its own welfare by tariffing a small partner whose goods are easily replaced by others'.** This happens not because the tariff fails to divert expenditure, but because the diversion is so complete that the price of the replacements rises. By Proposition 5.2, the same country gains from a uniform tariff at the same baseline. In the "+" cells of Table 2, $g_B < 0$ beyond the root, so $g_C > 0$ there: tariffing the bystander instead would pay. **The gains from protection are guaranteed only for uniform protection; discrimination can backfire.** The two-root case requires richer network asymmetry: in a representative example, $L_B/L_C = 2$, $\beta_{AB} = 0.2$, $\beta_{BA} = 0.9$ and $\beta_{CA} = 0.1$, the condition fails for $6.29 < \rho < 15.03$.

Table 3: Bystander reversal, NEER depreciation and aggregate terms of trade under a unilateral tariff, $N = 3$

w_{AC}	L_B	$d\nu_C > 0$ for $\rho >$	$d\nu_A^E > 0$ for $\rho >$	$R_{A\|B} > 0$ for $\rho >$
0.75	0.49	2.72	4.98	3.97
0.85	0.22	2.59	3.97	3.40
0.90	0.12	2.52	3.66	3.20

Notes: US-like parameters common to all countries, $\beta_{AB} = 0.2$, all other primitives symmetric. L_B is set so that A's baseline trade weight on C equals w_{AC} at $\rho = 3\rho_D$. Primitives are then held fixed and the free-trade baseline is re-solved at each ρ. $d\nu_A^E$ is the trade-weighted NEER; $R_{A|B}$ is the import-weighted aggregate terms of trade of Eq. (5.1). MML holds for all $\rho \leq 20$.

The trade-weighted NEER. The same logic applies to the NEER with trade weights. When the bystander carries most of A's trade, the NEER can depreciate under a *unilateral* tariff. Table 3 fixes $\beta_{AB} = 0.2$ and sets L_B so that w_{AC} takes the stated value. Its last column shows that the trade-weighted and import-weighted thresholds differ substantially in the same economy. In the second row, whose primitives are nearly those of example (b), the NEER depreciates only for $\rho > 3.97$, whereas the aggregate terms of trade deteriorate for $\rho > 3.40$ (3.34 at the fixed baseline of example (b)). The gap is a matter of weighting, not of method. A buys little from B but B buys a good deal from A, so the target's weight in A's trade ($w_{AB} = 0.15$) exceeds its weight in A's imports ($b_{AB} = 0.095$). The favorable move against the target therefore counts for more in the NEER than in the welfare-relevant terms of trade.

Other casualties of asymmetry (Section C).

- *Thresholds move.* Near the symmetric point the bystander threshold rises with the target's import weight ($d\rho^*/d\beta_{AB} = +3.70$), with its size ($+0.99$) and with A's home bias ($+5.31$), and falls slightly with bystander size (-0.20), at the calibration $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$ of Fig. 2; the US-like values have the same signs (Table 6).
- *Thresholds can disappear.* With primitives rather than shares held fixed, a higher ρ also shifts baseline shares toward the cheapest origin, and a large target can crowd the bystander out of A's imports. At the same calibration, with $L_B = 2.2$ the bystander reverses only for $5.99 < \rho < 53.8$, and with $L_B \geq 2.6$ it never reverses for $\rho \leq 1000$.
- *The ordering $d\nu_B < d\nu_C$ can fail*, but in every case found only with $\rho < \eta$.

5.4 Heterogeneous bystanders

With asymmetric bystanders the composite of Proposition 3.4 no longer exists. Partition the system into the target and the remaining countries R. With q denoting A's import weights within R,

$$d\nu_C^A \equiv q' d\nu_R = -q' J_{RR}^{-1} J_{RB}\, d\nu_B - q' J_{RR}^{-1} F_R\, d\tau, \tag{5.8}$$

which depends on the full internal adjustment structure of the bloc. The aggregate terms-of-trade condition still has three-country form, by the definition of $\bar{R}_{-B}^{B}$, but the bloc cannot generally be represented by a country with primitive parameters. Reversal and the failure of aggregation share an origin. The nested structure that lets ρ direct expenditure toward other foreign suppliers is exactly what makes it matter *which* suppliers receive it. With a single, unnested elasticity the bloc aggregates exactly, but bystander reversal disappears.

5.5 A stylized reading of the 2025 episode

We do not calibrate. A static, long-run, balanced-trade model compared against monthly exchange-rate data is not a test, and Section 5.3 shows that the answer under realistic asymmetry depends on network details that a three-country stylization cannot capture. We therefore ask only which way each feature of the 2025 episode pushes the trade channel, and we apply the same standard in both directions: a movement is *consistent with* the trade channel when the model predicts its sign in the relevant configuration, and *not explained by* the trade channel when the model predicts the opposite sign for every admissible ρ. Neither statement attributes the movement to its cause. The rule applies to a phase only insofar as the phase approximates a single configuration of the model, and in 2025 none did cleanly: each phase mixed discriminatory, retaliatory and near-uniform elements. Figure 7 shows the exchange rates and the tariff timeline, and Table 4 reports the change within each phase. Exchange-rate changes are measured relative to the January 2025 average, before the first tariff order of February 1, and signed so that a positive number is a dollar depreciation. The model's effective exchange rate corresponds to the full broad dollar index, which includes China; the index excluding China isolates the bystanders.

Phase 1: unilateral tariffs on China (February 1–April 2). The United States raised tariffs on China by 20 percentage points in two steps, effective February 4 and March 4. China's countermeasures in this phase were product-specific rather than across the board, and there was no general US tariff on third countries. The phase was not a clean unilateral experiment, however. US tariffs of 25 percent on Canada and Mexico took effect on March 4, and Canada retaliated the same day with a 25 percent surtax on C$30 billion of US goods

Table 4: Exchange-rate changes within each phase of the 2025 episode (%, positive = dollar depreciation)

	Unilateral Feb 1–Apr 2		Retaliation Apr 3–May 12		Near-uniform Aug 7–Dec 31	
	Monthly	Daily	Monthly	Daily	Monthly	Daily
EUR	4.4	4.5	3.9	2.2	0.3	0.8
CNY	0.6	−0.3	−0.7	0.9	1.9	2.7
CAD	0.2	1.0	2.7	2.2	−0.7	0.2
MXN	1.6	0.6	1.0	4.5	3.4	3.5
VND	−1.8	−2.1[a]	−0.5	−0.5[a]	0.0	0.5[a]
Broad index	2.1	1.7	1.7	2.3	−0.1	0.6
Broad index ex. China	2.2	1.9	2.0	2.5	−0.3	0.4

Notes: "Daily": from the last close before the phase to the last close in it. "Monthly": from the average of the month before the phase to the average of its last full month (January to March, March to April, July to December). USD/EUR, USD/CNY, USD/CAD and USD/MXN are H.10 noon rates; the broad index is the Federal Reserve's nominal broad dollar index, and the index excluding China removes the renminbi using its H.10 weight. [a]No daily VND series is available; BIS end-of-month rates over the same months. Sources as in Fig. 7.

(Department of Finance Canada, 2025); Canada and Mexico together carry 27.6 percent of the broad index. The configuration mixed a unilateral, discriminatory tariff on China with a retaliatory leg in North America. The US–China leg behaved as predicted: the dollar was roughly unchanged against the renminbi (−0.3 percent on daily data, +0.6 percent on monthly averages), with no sign of a depreciation against the target. Against bystanders the dollar weakened, by 4.5 percent against the euro and 1.9 percent against the broad index excluding China from the end of January to April 2, which is consistent with the trade channel if foreign suppliers are close substitutes ($\rho > N\rho_D$). Against the full broad index, which includes China, it weakened by 1.7 percent. A purely unilateral tariff predicts effective appreciation, but retaliation predicts effective depreciation once $\rho > \rho_D$ (Proposition 4.1), so in this mixed phase the effective move is not a test of either. Nearly all of the phase's effective movement came in the week of March 3–7 (1.8 percent against the broad index), which combined the North American tariffs and Canadian retaliation with the announcement by Germany's prospective governing parties of a large defence and infrastructure package and a loosening of the debt brake (Ripper, 2025). In that week the dollar fell by 4.4 percent against the euro, 1.3 percent against the peso and 0.3 percent against the Canadian dollar. The data do not separate these forces.

Phase 2: retaliation (April 3–May 12). China retaliated nearly one-for-one (US 145 percent, China 125 percent), while the United States imposed only a 10 percent baseline

tariff on others. The configuration moves toward bilateral retaliation, and the bystander threshold falls from about $N\rho_D$ to ρ_D:

- *Retaliation points to dollar depreciation against bystanders.* At US-like parameters $\rho_D \approx 1.3$, below almost all cross-origin estimates. The model predicts a roughly stable USD/RMB rate together with dollar weakness against bystanders and in effective terms. That is the pattern of the phase: the dollar moved little against the renminbi (+0.9 percent from April 2 to May 12, −0.7 percent between the March and April averages), but fell by 2.2–3.9 percent against the euro and by 2.0–2.5 percent against the broad index excluding China (Table 4). It is also the event-study regularity of Ostry et al. (2025).

- *Expected retaliation works the same way, as a matter of interpretation.* Our model is static, so expectations enter only through the tariff structure agents expect to persist. If markets price the long-run configuration as retaliatory, the relevant comparative static is Proposition 4.1, not Proposition 3.2. In a forward-looking extension, credible threats of retaliation would then move exchange rates before retaliation occurs. This is an interpretation, not a result of the model.

- *High cross-origin substitutability points the same way.* Trade between China and other suppliers, especially Vietnam, Mexico and other Asian producers, reallocated quickly (Fajgelbaum et al., 2024; Freund et al., 2024; Alfaro and Chor, 2023). That is the high-ρ environment in which diversion is strong.

- *China's size does not.* China is large in absolute terms and a substantial US supplier. Near the symmetric point, a larger target and a larger import weight on the target both *raise* the bystander threshold (Section 5.3). They also move the aggregate terms-of-trade condition toward its conventional sign.[1] The configuration is therefore not clear-cut. Retaliation and ρ lean toward dollar depreciation against bystanders, while China's weight leans against it. Where the balance falls depends on bilateral shares and sizes that a three-country stylization summarizes too crudely.

[1] Placed in the grid of Table 2 with 2024 data (US imports by origin from the US Census Bureau, China's from its customs statistics, the rest of the world's trade with both as residuals, and sizes from PPP GDP), a US–China–rest-of-world configuration falls in the region where $q_2 > 0$. In a three-country network calibrated to the observed import-origin shares, the aggregate terms-of-trade condition for a US tariff on China fails only for ρ above about 4.5–4.7, while the bilateral bystander threshold is about 3.9–4.0; the ranges span a common US-like home bias and the measured US home share. The bystander threshold lies at the upper end of the central range of estimates, $\rho \in [2, 4]$, and the aggregate threshold just above it. The calibration imposes a balanced baseline, ignores gaps between mirror trade statistics and uses model-implied home shares for China and the rest of the world. It places the episode on the map; it is not a calibration of exchange rates.

Phase 3: near-uniform tariffs (August 7–December 31). After the Geneva truce (effective May 14) the US tariff on China was 30 percent, falling to 20 percent on November 10, and China kept a 10 percent tariff on all US goods. Under the framework agreements in force from August 7, US tariffs were 15 percent on the EU and 10–20 percent on most others. Relative treatment had narrowed but not disappeared. Until November 10, China faced 30 percent against 10–20 percent for most others, and China's retaliation remained in place; in North America, the US rate on non-USMCA Canadian goods rose to 35 percent on August 1, and Canada's countermeasures on steel, aluminium and autos stayed in force. For the uniform component the relevant experiment is average protection, which predicts mild dollar appreciation in effective terms (Proposition 5.2). The residual tariff gaps, met by retaliation, are retaliation configurations in miniature, and they lean toward effective depreciation if $\rho > \rho_D$. Within the phase the dollar showed no sign of the predicted appreciation: the broad index barely moved (-0.1 to $+0.6$ percent, Table 4). In levels, the dollar in December was 13.1 percent weaker against the euro, 7.2 percent weaker against the broad index and 7.7 percent weaker against the index excluding China than in January. That level sums all phases, including retaliation, and most of it built up before August: from Geneva to August 6 alone the dollar fell by a further 4.9 percent against the euro and 2.5 percent against the broad index. Forces outside the model that plausibly matter for the level include risk premia and tariff uncertainty, the erosion of convenience yields, valuation effects, monetary policy and expectations of renewed escalation.

Vietnam illustrates the classification point. The dollar strengthened against the dong through 2025, by 3.2 percent between January and December, nearly all of it by July, which looks inconsistent with a bystander-depreciation story. But Vietnam was itself a principal target: a 46 percent rate was announced in April, and a 20 percent rate applied from August 7. From August 7 to November 10, Vietnam's 20 percent against China's 30 percent was relative treatment in Vietnam's favour, a bystander-like position in which strong diversion leans toward dollar weakness against the dong. Only after November 10, when both faced 20 percent, was there little relative treatment between them, so that the model predicts the conventional own-tariff result. Within the near-uniform phase the rate was roughly flat (0.0 to $+0.5$ percent).

What a quantitative general trade model would need. The analysis says what to look for in richer models. Retaliation lowers the threshold for multilateral depreciation. High cross-origin substitutability raises the diversion force. Large, heavily weighted targets work against both. Near-uniform tariffs collapse the problem back to average protection.

Quantitative multi-country models with input–output linkages (e.g., Ignatenko et al., 2025; Darracq Pariès et al., 2026) can contain all of these features. Our results identify which combinations of them determine the sign.

6 Conclusion

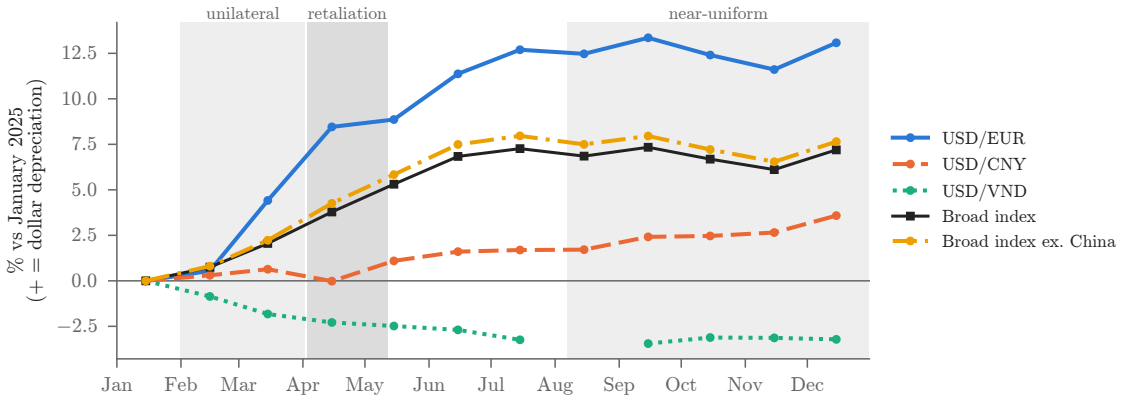
In a two-country world, the bilateral exchange rate, the effective exchange rate and the terms of trade coincide, and a tariff appreciates the tariffing country's currency wherever Marshall–Lerner holds. In a multilateral world they separate. We have characterized what the real trade channel implies once they do.

The general response is the inverse of a multilateral adjustment matrix applied to a vector of tariff disturbances. A multilateral Marshall–Lerner condition makes adjustment well behaved without fixing the sign of any bilateral response. Under symmetry, average protection moves the effective rate and relative treatment moves bilateral rates around it. In a nested-CES model the conventional result survives exactly at the effective level under unilateral tariffs. It fails bilaterally against bystanders when cross-origin substitution exceeds N times the domestic diversion margin. An exact three-country representation shows that the role of N comes from the inward-looking, highly substitutable trade of the bystander bloc, not from its size. Under bilateral retaliation the belligerents' bilateral rate is unchanged, and once cross-origin substitution exceeds the domestic diversion margin itself each belligerent depreciates against bystanders and in effective terms; only in a multilateral tariff war does the tariffing country depreciate against everyone. Under asymmetry, even the effective and welfare-relevant conclusions can fail for discriminatory tariffs: tariffing a small, lightly weighted partner whose goods are easily replaced can worsen the tariffing country's aggregate terms of trade. A uniform tariff, by contrast, appreciates the currency against every partner and raises welfare at any baseline under MML. The gains from protection are guaranteed only for uniform protection.

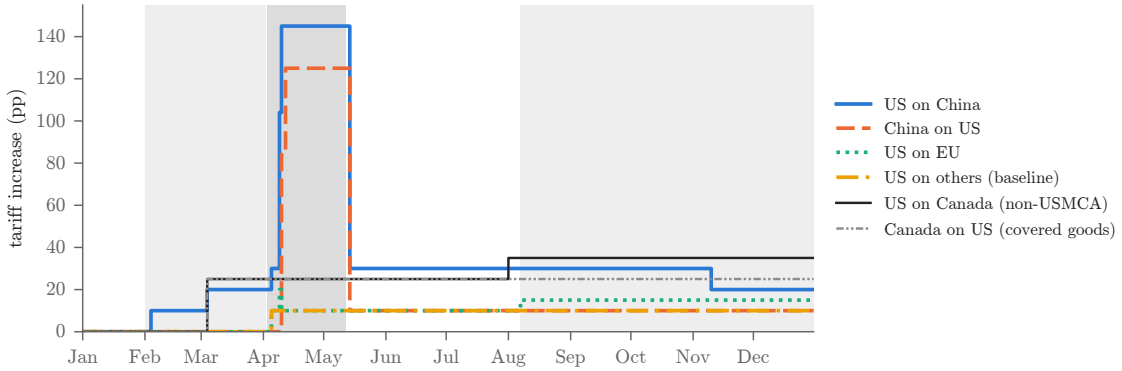
The implications run both ways. Appreciation against a tariffed partner is not evidence of broad currency strength. A multilateral depreciation during a discriminatory or retaliatory tariff episode can be consistent with the trade channel, so the two-country prediction alone does not show that financial forces were at work. Conversely, in a unilateral or nearly uniform configuration the trade channel predicts effective appreciation, so effective weakness in such a configuration is not explained by it. Whether an episode is such a configuration has to be checked phase by phase, because actual episodes mix unilateral, retaliatory and near-uniform elements.

Natural next steps are to embed the decomposition in a quantitative multi-country model

with input–output linkages and heterogeneous elasticities, and to add nominal rigidities, invoicing and a financial account. This would let the long-run real benchmark derived here interact with the short-run monetary and portfolio dynamics that dominate exchange rates at business-cycle frequencies.



(a) Exchange rates, % deviation from the January 2025 average (positive = dollar depreciation)



(b) Headline tariff increases above the pre-2025 baseline (percentage points)

Figure 7: The 2025 episode through the lens of the decomposition (stylized). Shading marks the unilateral phase (February 1–April 2), the retaliation phase (April 3–May 12, to the Geneva statement) and the near-uniform phase (framework rates in force from August 7). (a) Monthly averages of USD per EUR, CNY and VND, the Federal Reserve H.10 nominal broad dollar index, and the broad index with the renminbi removed using its published H.10 weight and renormalized (both indices inverted so that a rise is a dollar depreciation), all relative to the January 2025 average, before the first tariff order. The August VND average is missing in the source. (b) Effective dates of headline rates: the US tariff on China, China's across-the-board tariff on the United States (China's product-specific countermeasures of February and March are not shown), the US tariff on the EU (the August rate is the 15 percent all-in rate) the US baseline on other partners, the US tariff on Canada (from March 7 on non-USMCA goods only; the rate on Mexico is the same until August 1 and 25 percent thereafter) and Canada's 25 percent surtax on covered US goods (coverage widened on March 13 and April 9 and narrowed to steel, aluminium and autos on September 1). No model magnitudes are plotted. Sources: FRED (DEXUSEU, DEXCHUS, DEXCAUS, DEXMXUS, DTWEXBGS), Federal Reserve H.10 weights, BIS (VND), Federal Register, Department of Finance Canada and China's State Council Tariff Commission announcements.

References

Acharya, Viral V. and Toomas Laarits, "Tariff War Shock and the Convenience Yield of US Treasuries—A Hedging Perspective," NBER Working Paper 34640, National Bureau of Economic Research 2026.

Alfaro, Laura and Davin Chor, "Global Supply Chains: The Looming 'Great Reallocation'," in "Structural Shifts in the Global Economy: Jackson Hole Economic Policy Symposium," Kansas City, MO: Federal Reserve Bank of Kansas City, 2023. Also NBER Working Paper 31661.

Arrow, Kenneth J. and Leonid Hurwicz, "On the Stability of the Competitive Equilibrium, I," *Econometrica*, 1958, *26* (4), 522–552.

_ , H. D. Block, and Leonid Hurwicz, "On the Stability of the Competitive Equilibrium, II," *Econometrica*, 1959, *27* (1), 82–109.

Artus, Jacques R. and Rudolf R. Rhomberg, "A Multilateral Exchange Rate Model," *IMF Staff Papers*, 1973, *20* (3), 591–611.

Auray, Stéphane, Michael B. Devereux, and Aurélien Eyquem, "Trade Wars, Nominal Rigidities, and Monetary Policy," *Review of Economic Studies*, 2025, *92* (4), 2228–2270.

Bagwell, Kyle and Robert W. Staiger, "An Economic Theory of GATT," *American Economic Review*, 1999, *89* (1), 215–248.

_ and _ , *The Economics of the World Trading System*, Cambridge, MA: MIT Press, 2002.

Barattieri, Alessandro, Matteo Cacciatore, and Fabio Ghironi, "Protectionism and the Business Cycle," *Journal of International Economics*, 2021, *129*, 103417.

Bergin, Paul R. and Giancarlo Corsetti, "The Macroeconomic Stabilization of Tariff Shocks: What Is the Optimal Monetary Response?," *Journal of International Economics*, 2023, *143*, 103758.

Bianchi, Javier and Louphou Coulibaly, "The Optimal Monetary Policy Response to Tariffs," NBER Working Paper 33560, National Bureau of Economic Research 2025.

Boehm, Christoph E., Andrei A. Levchenko, and Nitya Pandalai-Nayar, "The Long and Short (Run) of Trade Elasticities," *American Economic Review*, 2023, *113* (4), 861–905.

Bond, Eric W. and Constantinos Syropoulos, "The Size of Trading Blocs: Market Power and World Welfare Effects," *Journal of International Economics*, 1996, *40* (3–4), 411–437.

Broda, Christian and David E. Weinstein, "Globalization and the Gains from Variety," *Quarterly Journal of Economics*, 2006, *121* (2), 541–585.

_ , **Nuno Limão, and David E. Weinstein**, "Optimal Tariffs and Market Power: The Evidence," *American Economic Review*, 2008, *98* (5), 2032–2065.

Cardani, Roberta, Michael Kühl, Jemima Peppel-Srebrny, and Rolf Strauch, "Exchange Rate Uncertainty, Tariff Hikes, and Adjustment Costs," VoxEU.org column, 12 May 2025.

Chang, Won and L. Alan Winters, "How Regional Blocs Affect Excluded Countries: The Price Effects of MERCOSUR," *American Economic Review*, 2002, *92* (4), 889–904.

Costinot, Arnaud and Iván Werning, "Lerner Symmetry: A Modern Treatment," *American Economic Review: Insights*, 2019, *1* (1), 13–26.

_ , **Dave Donaldson, Jonathan Vogel, and Iván Werning**, "Comparative Advantage and Optimal Trade Policy," *Quarterly Journal of Economics*, 2015, *130* (2), 659–702.

Department of Finance Canada, "Canada Announces Robust Tariff Package in Response to Unjustified U.S. Tariffs," News release, March 4. https://www.canada.ca/en/department-finance/news/2025/03/canada-announces-robust-tariff-package-in-response-to-unjustified-us-tariffs.html 2025.

Dornbusch, Rudiger, "Tariffs and Nontraded Goods," *Journal of International Economics*, 1974, *4* (2), 177–185.

Eichengreen, Barry J., "A Dynamic Model of Tariffs, Output and Employment under Flexible Exchange Rates," *Journal of International Economics*, 1981, *11* (3), 341–359.

Erceg, Christopher, Andrea Prestipino, and Andrea Raffo, "Trade Policies and Fiscal Devaluations," *American Economic Journal: Macroeconomics*, 2023, *15* (4), 104–140.

Fajgelbaum, Pablo D. and Amit K. Khandelwal, "Tariffs in 2025: Short-Run Impacts on the U.S. Economy," *Brookings Papers on Economic Activity*, 2026. Spring 2026 conference draft; also NBER Working Paper 35064.

Fajgelbaum, Pablo, Pinelopi Goldberg, Patrick Kennedy, Amit Khandelwal, and Daria Taglioni, "The US-China Trade War and Global Reallocations," *American Economic Review: Insights*, 2024, *6* (2), 295–312.

Feenstra, Robert C., Philip Luck, Maurice Obstfeld, and Katheryn N. Russ, "In Search of the Armington Elasticity," *Review of Economics and Statistics*, 2018, *100* (1), 135–150.

Freund, Caroline, Aaditya Mattoo, Alen Mulabdic, and Michele Ruta, "Is US Trade Policy Reshaping Global Supply Chains?," *Journal of International Economics*, 2024, *152*, 104011.

Gopinath, Gita, Emine Boz, Camila Casas, Federico J. Díez, Pierre-Olivier Gourinchas, and Mikkel Plagborg-Møller, "Dominant Currency Paradigm," *American Economic Review*, 2020, *110* (3), 677–719.

Hicks, John R., *Value and Capital: An Inquiry into Some Fundamental Principles of Economic Theory*, Oxford: Clarendon Press, 1939.

Ignatenko, Anna, Ahmad Lashkaripour, Luca Macedoni, and Ina Simonovska, "Making America Great Again? The Economic Impacts of Liberation Day Tariffs," *Journal of International Economics*, 2025, *157*, 104138.

Itskhoki, Oleg and Dmitry Mukhin, "The Optimal Macro Tariff," NBER Working Paper 33839, National Bureau of Economic Research 2025.

Jeanne, Olivier, "Currency Wars, Trade Wars, and Global Demand," NBER Working Paper 29603, National Bureau of Economic Research 2021.

Jiang, Zhengyang, Arvind Krishnamurthy, Hanno Lustig, Robert Richmond, and Chenzi Xu, "Dollar Upheaval: This Time Is Different," Research Paper 4241, Stanford University Graduate School of Business 2025.

Johnson, Harry G., "Optimum Tariffs and Retaliation," *Review of Economic Studies*, 1953, *21* (2), 142–153.

Kalemli-Özcan, Şebnem, Can Soylu, and Muhammed A. Yıldırım, "Global Trade, Tariff Uncertainty, and the US Dollar," *AEA Papers and Proceedings*, 2026, *116*, 47–52. Longer version: "Tariff Uncertainty and the U.S. Dollar," NBER Working Paper 34728, 2026; also CEPR Discussion Paper 21062.

Kemp, Murray C. and Henry Y. Wan Jr., "An Elementary Proposition Concerning the Formation of Customs Unions," *Journal of International Economics*, 1976, *6* (1), 95–97.

Krugman, Paul, "The Macroeconomics of Protection with a Floating Exchange Rate," *Carnegie-Rochester Conference Series on Public Policy*, 1982, *16*, 141–182.

_ , "Is Bilateralism Bad?," in Elhanan Helpman and Assaf Razin, eds., *International Trade and Trade Policy*, Cambridge, MA: MIT Press, 1991, pp. 9–23.

Lerner, Abba P., "The Symmetry between Import and Export Taxes," *Economica*, 1936, *3* (11), 306–313.

Li, Wei, Bin Qiu, Zi Wang, Yanbo Zeng, and Yuan Zi, "Armington Elasticities and the Third-Country Effects of Trade Wars," CEPR Discussion Paper 17409, Centre for Economic Policy Research 2022.

Lindé, Jesper and Andrea Pescatori, "The Macroeconomic Effects of Trade Tariffs: Revisiting the Lerner Symmetry Result," *Journal of International Money and Finance*, 2019, *95*, 52–69.

Metzler, Lloyd A., "Stability of Multiple Markets: The Hicks Conditions," *Econometrica*, 1945, *13* (4), 277–292.

_ , "Tariffs, the Terms of Trade, and the Distribution of National Income," *Journal of Political Economy*, 1949, *57* (1), 1–29.

Mosak, Jacob L., *General-Equilibrium Theory in International Trade* number 7. In 'Cowles Commission Monograph.', Bloomington, IN: Principia Press, 1944.

Mundell, Robert A., "Flexible Exchange Rates and Employment Policy," *Canadian Journal of Economics and Political Science*, 1961, *27* (4), 509–517.

_ , "Tariff Preferences and the Terms of Trade," *Manchester School of Economic and Social Studies*, 1964, *32* (1), 1–13.

Ornelas, Emanuel, "Trade Creating Free Trade Areas and the Undermining of Multilateralism," *European Economic Review*, 2005, *49* (7), 1717–1735.

Ostry, Daniel, Simon Lloyd, and Giancarlo Corsetti, "Trading Blows: The Exchange-Rate Response to Tariffs and Retaliations," Staff Working Paper 1139, Bank of England 2025. Also CEPR Discussion Paper 20452.

Pariès, Matthieu Darracq, Aurélien Eyquem, and Valentin Jouvanceau, "The Third-Country Effects of Trade Wars," ECB Working Paper 3213, European Central Bank 2026.

Ripper, Kirsten, "CDU's Friedrich Merz Secures Billion-Euro Deal for Germany's Military and Infrastructure," Euronews, March 4. https://www.euronews.com/my-europe/2025/03/04/cdus-friedrich-merz-secures-billion-euro-deal-for-germanys-military-and-infrastructure 2025.

Viner, Jacob, *The Customs Union Issue*, New York: Carnegie Endowment for International Peace, 1950.

Werning, Iván, Guido Lorenzoni, and Veronica Guerrieri, "Tariffs as Cost-Push Shocks: Implications for Optimal Monetary Policy," NBER Working Paper 33772, National Bureau of Economic Research 2025.

Yi, Sang-Seung, "Free-Trade Areas and Welfare: An Equilibrium Analysis," *Review of International Economics*, 2000, *8* (2), 336–347.

A The Linearized System

At a balanced free-trade baseline, let b_{ij} be origin shares within i's import bundle, h_i home shares of i's tradable spending, $y_i \equiv \tilde{I}_i$ incomes in the common currency, $s_{ij} = \alpha_T(1-h_i)b_{ij}$ the share of i's income spent on j's good, and $f_{ij} = s_{ij}\tilde{I}_i$; at the symmetric point $h_i = \alpha_D$ and $m = \alpha_T(1-\alpha_D)$. Log-differentiating the demand system, Eq. (3.2) and Eq. (3.3), with η (and, if desired, ρ) allowed to differ by country,

$$\begin{aligned}
dp_{ij} &= d\nu_j - d\nu_i + d\tau_{ij}, \qquad dp_{ii} = 0, \\
d\log P_{Mi} &= \textstyle\sum_{j\neq i} b_{ij}\, dp_{ij}, \qquad d\log P_{Ti} = (1-h_i)\, d\log P_{Mi}, \\
d\log s_{ii} &= -(1-\eta)(1-h_i)\, d\log P_{Mi}, \\
d\log s_{ij} &= (1-\rho)(dp_{ij} - d\log P_{Mi}) + (1-\eta)h_i\, d\log P_{Mi}, \\
d\log \tilde{I}_i &= d\nu_i + \textstyle\sum_{j\neq i} s_{ij}\, d\tau_{ij}, \\
d\mathrm{TB}_i &= \textstyle\sum_{k\neq i} f_{ki}(d\log s_{ki} + d\log \tilde{I}_k - d\tau_{ki}) - \sum_{j\neq i} f_{ij}(d\log s_{ij} + d\log \tilde{I}_i - d\tau_{ij}).
\end{aligned} \tag{A.1}$$

The income equation uses that at free trade $d[\tau_{ij}/(1+\tau_{ij})] = d\tau_{ij}$ and that changes in realized shares are multiplied by zero tariffs, so only the direct revenue term survives. These equations determine the matrices J and G of Eq. (2.2) at any free-trade baseline, as general functions of (y_i, h_i, b_{ij}), the elasticities and an arbitrary matrix of tariff changes. In the replication code they are implemented directly and checked against central finite differences of the nonlinear N-country solver at tariffs of $\pm 10^{-4}$, with agreement to better than 10^{-6} at symmetric and asymmetric baselines.

Lemma A.1 (Jacobian at the symmetric baseline)**.** *At the symmetric balanced baseline, where $b_{ij} = 1/(N-1)$ and $f_{ij} = m/(N-1)$, the full Jacobian is $\tilde{J}_{il} = mM_N/(N-1)$ for $l \neq i$ and $\tilde{J}_{ii} = -mM_N$. Deleting A gives*

$$-J = c\,(NI - \mathbf{1}\mathbf{1}'), \qquad c \equiv \frac{mM_N}{N-1}, \qquad j_S = c, \qquad j_D = Nc. \tag{A.2}$$

Proof. Set $d\tau = 0$, so $dp_{ij} = d\nu_j - d\nu_i$ and $d\log \tilde{I}_i = d\nu_i$. Writing $\Sigma \equiv \sum_j d\nu_j$,

$$d\log P_{Mk} = \frac{\Sigma - N\, d\nu_k}{N-1}, \qquad \frac{\partial \log P_{Mk}}{\partial \nu_l} = \frac{1 - N\delta_{kl}}{N-1}.$$

On the import side $\sum_{j\neq i}(dp_{ij} - d\log P_{Mi}) = 0$, so the $(1-\rho)$ terms cancel when $d\log s_{ij}$ is summed over $j \neq i$, leaving $\sum_{j\neq i} d\log s_{ij} = (N-1)(1-\eta)\alpha_D\, d\log P_{Mi}$. Substituting into

the trade-balance equation,

$$d\mathrm{TB}_i = \frac{m}{N-1}\Big[\sum_{k\neq i}(d\log s_{ki} + d\nu_k) - (N-1)(1-\eta)\alpha_D\, d\log P_{Mi} - (N-1)\, d\nu_i\Big].$$

Fix $l \neq i$ and differentiate term by term:

$$\frac{\partial}{\partial \nu_l}\sum_{k\neq i}\log s_{ki} = -(1-\rho)\frac{N-2}{N-1} - \frac{(1-\eta)\alpha_D}{N-1}.$$

The export income terms contribute one, the import-side price-index term contributes $-(1-\eta)\alpha_D$, and the import income term contributes zero. Collecting terms,

$$\tilde{J}_{il} = \frac{m}{(N-1)^2}\Big[(N-1) - (1-\rho)(N-2) + N\alpha_D(\eta-1)\Big] = \frac{mM_N}{N-1}.$$

The diagonal follows from homogeneity, $\tilde{J}_{ii} = -(N-1)\tilde{J}_{il} = -mM_N$. Deleting row and column A leaves diagonal entries $c(N-1)$ and off-diagonal entries $-c$, which is Eq. (A.2); its eigenvalues are c on $\mathrm{span}\{\mathbf{1}\}$ and Nc on the zero-sum subspace. □

The tariff disturbance. At unchanged exchange rates only partner i's exports to A are directly affected by a change in A's tariffs t, so

$$d\mathrm{TB}_i\Big|_{\nu} = f_{Ai}\Big(d\log s_{Ai} + d\log \tilde{I}_A - t_i\Big) = \frac{m}{N-1}\Big[-\rho_D\,\bar{t} - \rho\,(t_i - \bar{t})\Big]. \tag{A.3}$$

Indeed, $d\log P_{MA} = \bar{t}$, $d\log s_{Ai} = (1-\rho)(t_i - \bar{t}) + (1-\eta)\alpha_D\bar{t}$ and $d\log \tilde{I}_A = m\bar{t}$, and $1 - (1-\eta)\alpha_D - m = \rho_D$. Reading off the coefficients on $\bar{t}$ and $\tilde{t}_i$,

$$g_S = -\frac{m\rho_D}{N-1}, \qquad g_D = -\frac{m\rho}{N-1}, \tag{A.4}$$

and dividing by the eigenvalues in Eq. (A.2) gives $\lambda_A = -\rho_D/M_N$ and $\lambda_R = -\rho/(NM_N)$, the coefficients of Theorem 3.1.

The two-country case. At $N = 2$ each country has a single import origin, so $dp_{AB} = d\log P_{MA} = d\nu_B + d\tau$ and $dp_{BA} = d\log P_{MB} = -d\nu_B$. Then

$$d\log s_{AB} = (1-\eta)\alpha_D(d\nu_B + d\tau), \qquad d\log s_{BA} = -(1-\eta)\alpha_D\, d\nu_B,$$
$$d\log \tilde{I}_A = m\, d\tau, \qquad d\log \tilde{I}_B = d\nu_B.$$

Substituting into the trade balance, and using that the tariff wedge lowers the border value of A's imports by $d\tau$,

$$d\mathrm{TB}_A = m\Big[\big(1+2\alpha_D(\eta-1)\big)d\nu_B + \big(1-m+\alpha_D(\eta-1)\big)d\tau\Big],$$

which is the two-country system with $M_2 = 1+2\alpha_D(\eta-1)$ and $\rho_D = 1-m+\alpha_D(\eta-1) = \alpha_D\eta + (1-\alpha_D)(1-\alpha_T)$; Eq. (3.4) follows.

For the elasticity form, hold income fixed and let $M(p)$ be import volume at consumer price $p = p_{AB}$. Since import value is $s_{AB}I_A$,

$$\varepsilon_M \equiv -\frac{\partial \log M}{\partial \log p} = 1 - \frac{\partial \log s_{AB}}{\partial \log p} = 1+\alpha_D(\eta-1),$$

and by symmetry $\varepsilon_X = 1+\alpha_D(\eta-1)$. Differentiating $\mathrm{TB}_A(e,\tau) = X(1/e) - e\,M((1+\tau)e)$ at $\tau = 0$ and using balanced trade, $X = eM$, gives $\partial \mathrm{TB}_A/\partial e = M(\varepsilon_X + \varepsilon_M - 1)$. Hence $\varepsilon_X + \varepsilon_M - 1 = M_2$, and the stability condition $M_2 > 0$ is exactly $\varepsilon_X + \varepsilon_M > 1$.

The unilateral tariff. For $t = d\tau\, e_B$, $\bar{t} = d\tau/(N-1)$, $\tilde{t}_B = \frac{N-2}{N-1}d\tau$ and $\tilde{t}_C = -\frac{d\tau}{N-1}$. By Theorem 3.1,

$$d\nu_B = \frac{\lambda_A + (N-2)\lambda_R}{N-1}\,d\tau = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N}\,d\tau,$$
$$d\nu_C = \frac{\lambda_A - \lambda_R}{N-1}\,d\tau = \frac{\rho - N\rho_D}{N(N-1)M_N}\,d\tau,$$

while $d\log e_{BC} = d\nu_C - d\nu_B = -\lambda_R\,d\tau$ and $d\nu_A^E = \lambda_A\,d\tau/(N-1)$. The bystander's disturbance follows from Eq. (A.3) with $t_C = 0$: $F_C = \frac{m}{N-1}\Big[-\frac{\rho_D}{N-1} + \frac{\rho}{N-1}\Big]$, which is Eq. (3.10).

The single-elasticity case. By the definition of ρ_D, $N\rho_D = N[1+\alpha_D(\eta-1)-\alpha_T(1-\alpha_D)]$, which rearranges to Eq. (3.9); both terms on its right-hand side are nonnegative for $\alpha_D \geq 1/N$, and equality requires $\alpha_D = 1/N$ and $\alpha_T = 1$. At $\alpha_D = 1/N$ with $\rho = \eta = \sigma$, $N\rho_D - \sigma = (N-1)(1-\alpha_T)$ and $M_N = \sigma$, so $d\nu_C/d\tau = -(1-\alpha_T)/(N\sigma)$.

Relative treatment across partners. With $\bar{t} > 0$, A depreciates against partner j if and only if

$$\frac{\rho}{N}(\bar{t} - t_j) > \rho_D\,\bar{t} \qquad \Longleftrightarrow \qquad \frac{t_j}{\bar{t}} < 1 - \frac{N\rho_D}{\rho}, \tag{A.5}$$

which follows by substituting λ_A and λ_R into $d\nu_j = \lambda_A\bar{t} + \lambda_R(t_j - \bar{t})$. The right-hand side is positive only if $\rho > N\rho_D$, so depreciation against a relatively lightly taxed partner requires both $\rho > N\rho_D$ and a sufficiently large tariff differential.

Tariffs on several partners. If A raises its tariff by $d\tau$ on a set of k partners, $1 \le k \le N-1$, then for every untargeted partner C and every targeted partner B,

$$\frac{d\log e_{AC}}{d\tau} = \frac{k(\rho - N\rho_D)}{N(N-1)M_N}, \qquad \frac{d\log e_{AB}}{d\tau} = -\frac{Nk\rho_D + (N-1-k)\rho}{N(N-1)M_N}, \qquad \frac{d\nu_A^E}{d\tau} = -\frac{k\rho_D}{(N-1)M_N}. \tag{A.6}$$

This follows by substituting $\bar{t} = k\,d\tau/(N-1)$, $\tilde{t}_B = (N-1-k)\,d\tau/(N-1)$ and $\tilde{t}_C = -k\,d\tau/(N-1)$ into Eq. (3.7). For every $k \le N-2$ the bystander response has the same threshold, $\rho > N\rho_D$.

Real exchange rates. From $q_{Aj} = e_{Aj}P_j/P_A$, with P_i the consumer price index in own currency and P_{Ni} fixed, $d\log q_{Aj} = d\nu_j + \alpha_T(1-\alpha_D)(d\log P_{Mj} - d\log P_{MA})$. For a tariff vector t imposed by A alone, symmetry gives $d\log P_{MA} = d\nu_A^E + \bar{t}$ and $d\log P_{Mj} = d\nu_A^E - \frac{N}{N-1}d\nu_j$ for a partner that imposes no tariff. Hence

$$d\log q_{Aj} = \Big(1 - \frac{mN}{N-1}\Big)d\nu_j - m\bar{t}, \qquad d\log q_{Aj} - dq_A^E = \Big(1 - \frac{mN}{N-1}\Big)\lambda_R\tilde{t}_j, \tag{A.7}$$

which with Eq. (3.7) gives Eq. (3.14). For a unilateral tariff, $d\log q_{Aj} = (1-\frac{mN}{N-1})d\nu_j - \frac{m}{N-1}d\tau$ for $j = B, C$, and averaging,

$$\frac{dq_A^E}{d\tau} = -\Big(1 - \frac{mN}{N-1}\Big)\frac{\rho_D}{(N-1)M_N} - \frac{m}{N-1}. \tag{A.8}$$

Three-country illustration. At $N = 3$, Eqs. (A.2) and (A.3) give the unilateral-tariff system explicitly:

$$J = -\frac{mM_3}{2}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad F = -\frac{m}{4}\begin{pmatrix} \rho + \rho_D \\ \rho_D - \rho \end{pmatrix}, \qquad M_3 = \frac{3\alpha_D(\eta-1) + \rho + 1}{2}. \tag{A.9}$$

With $M_3 > 0$,

$$\frac{d\log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_D}{6M_3} < 0, \qquad \frac{d\log e_{AC}}{d\tau} = \frac{\rho - 3\rho_D}{6M_3}, \qquad \frac{d\log e_{BC}}{d\tau} = \frac{\rho}{3M_3} > 0. \tag{A.10}$$

Thus A appreciates against B, B depreciates against C, and A depreciates against C if and only if $\rho > 3\rho_D$. By Eq. (2.4), $d\nu_C/d\tau \propto F_B + 2F_C \propto \rho - 3\rho_D$, so the sign of C's own disturbance F_C, which is positive once $\rho > \rho_D$, does not settle the sign of $d\nu_C$; Fig. 2a illustrates the intermediate case.

B Proofs

Throughout, $m \equiv \alpha_T(1-\alpha_D)$ and all derivatives are evaluated at a balanced free-trade baseline (Section A).

Proof of Lemma 2.3. Scaling every currency value by a common factor leaves all relative prices unchanged and scales every common-currency income, and hence every flow f_{ij} and trade balance TB_i, by the same factor, so TB_i is homogeneous of degree one in currency values. By Euler's theorem $\sum_l \tilde{J}_{il} = \mathrm{TB}_i$, which is zero at the balanced baseline. Summing Eq. (3.3) over i gives $\sum_i \mathrm{TB}_i \equiv 0$, so $\sum_i \tilde{J}_{il} = 0$. Under Assumption 2.1, $\tilde{J}_{il} \geq 0$ for $l \neq i$, so $\tilde{J}_{ii} = -\sum_{l\neq i} \tilde{J}_{il} \leq 0$. Deleting row and column A then gives, for each $i \neq A$,

$$|J_{ii}| - \sum_{l\neq i,A} J_{il} = \tilde{J}_{iA} \geq 0,$$

with strict inequality in every row for which $\tilde{J}_{iA} > 0$. Thus $-J$ is a Z-matrix that is weakly diagonally dominant, irreducible, and strictly diagonally dominant in at least one row. Its diagonal is strictly positive, since $|J_{ii}| = 0$ would force row i of J to vanish, contradicting irreducibility. Write $-J = D - B$ with $D = \mathrm{diag}(-J) > 0$ and $B \geq 0$. Every row sum of $D^{-1}B$ is at most one and at least one is strictly below one; with irreducibility, its spectral radius is $r < 1$. Hence $(-J)^{-1} = (I - D^{-1}B)^{-1}D^{-1} = \sum_{k\geq 0}(D^{-1}B)^k D^{-1} \geq 0$, nonsingularity follows from $r < 1$, and $\det(-J) = \det D \cdot \det(I - D^{-1}B) > 0$. □

Proof of Proposition 2.4. Let M be an $(N-1)\times(N-1)$ matrix with $PMP' = M$ for every permutation matrix P. Conjugating by the transposition of i and j gives $M_{ii} = M_{jj}$, and conjugating by a permutation that sends the ordered pair (i,j) to (k,l), $i \neq j$, $k \neq l$, gives $M_{ij} = M_{kl}$. Hence $M = aI + b\mathbf{1}\mathbf{1}'$. Since $\mathbf{1}'\mathbf{1} = N-1$ and $\mathbf{1}'v = 0$ for zero-sum v, $(aI + b\mathbf{1}\mathbf{1}')\mathbf{1} = (a + (N-1)b)\mathbf{1}$ and $(aI + b\mathbf{1}\mathbf{1}')v = av$. For $-J = a_J I + b_J \mathbf{1}\mathbf{1}'$ and $G = a_G I + b_G \mathbf{1}\mathbf{1}'$ this gives $j_S = a_J + (N-1)b_J$, $j_D = a_J$, $g_S = a_G + (N-1)b_G$ and $g_D = a_G$. For $j_S, j_D \neq 0$,

$$(-J)^{-1} = \frac{1}{j_D} I + \frac{1}{N-1}\left(\frac{1}{j_S} - \frac{1}{j_D}\right)\mathbf{1}\mathbf{1}',$$

as direct multiplication verifies. Thus $(-J)^{-1}$ and G share the common and zero-sum eigenspaces, with response coefficients $\lambda_A = g_S/j_S$ and $\lambda_R = g_D/j_D$; applying them to $t = \bar{t}\mathbf{1} + \tilde{t}$ gives Eq. (2.6). Because $\tilde{t}$ sums to zero, it does not affect the equally weighted average of the responses, which gives the first equality in Eq. (2.7); the second is immediate, and for any two partners $d\log e_{jk} = d\nu_k - d\nu_j = \lambda_R(\tilde{t}_k - \tilde{t}_j) = \lambda_R(t_k - t_j)$.

For the MML characterization, the off-diagonal entry of $-J$ is $b_J = (j_S - j_D)/(N-1)$, so $-J$ has the Z-matrix sign pattern if and only if $j_D \geq j_S$. Given $j_D \geq j_S > 0$, every entry

of $(-J)^{-1}$ above is nonnegative, so MML holds. Conversely, if $j_S \leq 0$, either $-J$ is singular or $(-J)^{-1}\mathbf{1} = \mathbf{1}/j_S$ is not nonnegative. □

Proof of Theorem 3.1. At the symmetric benchmark relabeling A's partners leaves the reduced system unchanged, so Proposition 2.4 applies. By Lemma A.1, $j_S = mM_N/(N-1)$ and $j_D = Nj_S$, so $j_S > 0$ if and only if $M_N > 0$, in which case $j_D > j_S$; by Proposition 2.4, MML is therefore equivalent to $M_N > 0$. If $\eta \geq 1$, every term in the bracket of Eq. (3.5) is nonnegative and the first is one, so $M_N > 0$. Given $M_N > 0$, dividing Eq. (A.4) by the eigenvalues in Eq. (A.2) gives $\lambda_A = -\rho_D/M_N$ and $\lambda_R = -\rho/(NM_N)$, and Eq. (3.7) is Eq. (2.6). □

Proof of Proposition 3.2. The three responses are derived in Section A. Under MML $M_N > 0$, and $\rho_D > 0$, so $d\nu_A^E < 0$ and $d\nu_B < 0$, and $d\nu_C > 0$ if and only if $\rho > N\rho_D$. Part (iv) is Eqs. (A.5) and (A.6). □

Proof of Corollary 3.3. See the single-elasticity paragraph of Section A. Since $N\rho_D \geq \eta$ when $\alpha_D \geq 1/N$, $\rho > N\rho_D$ implies $\rho > \eta$, and with $\rho = \eta$ the condition $\rho > N\rho_D$ cannot hold. □

Proof of Proposition 3.4. Label the countries A, B and bystanders $c = 1, \ldots, N-2$. Consider any tariff change in the class described in the proposition, and write the composite economy's variables with a capital C.

Step 1: bystanders move together. The baseline and the shock are invariant to permutations of the bystanders, and under MML the linear system Eq. (2.2) has a unique solution. That solution is therefore invariant to the same permutations, so $d\nu_c = d\nu_C$ for every bystander. The equilibrium is then characterized by two conditions, $d\mathrm{TB}_B = 0$ and $\sum_c d\mathrm{TB}_c = 0$ (each $d\mathrm{TB}_c$ is equal across bystanders), in the two unknowns $(d\nu_B, d\nu_C)$.

Step 2: the baseline flows coincide. In the N-country economy $f_{Ac} = f_{cA} = m/(N-1)$ for each bystander, so $\sum_c f_{Ac} = \sum_c f_{cA} = m(N-2)/(N-1)$. In the composite, $f_{AC} = \alpha_T(1-\alpha_D)\frac{N-2}{N-1} = m\frac{N-2}{N-1}$, and, since $1-h_C = 2(1-\alpha_D)/(N-1)$, $f_{CA} = \alpha_T(1-h_C)\cdot\frac{1}{2}\cdot(N-2) = m\frac{N-2}{N-1}$. The same holds for B, and $f_{AB} = f_{BA} = m/(N-1)$ in both economies. In particular the composite is balanced, and $s_{CA} = \alpha_T(1-h_C)/2 = m/(N-1) = s_{cA}$.

Step 3: A's and B's demand aggregate exactly. All bystander goods face the same price change in A, $dp_{Ac} = d\nu_C + d\tau_{Ac}$ with $d\tau_{Ac}$ common across c. Hence $d\log P_{MA} = \frac{1}{N-1}dp_{AB} + \frac{N-2}{N-1}dp_{Ac}$, which is the composite's index with $b_{AC} = \frac{N-2}{N-1}$, and $d\log s_{Ac} = (1-\rho)(dp_{Ac} - d\log P_{MA}) + (1-\eta)\alpha_D\, d\log P_{MA}$ is common across c and equal to the composite's $d\log s_{AC}$. A's income response, $d\log \tilde{I}_A = s_{AB}d\tau_{AB} + \sum_c s_{Ac}d\tau_{Ac}$, equals the composite's

because $\sum_c s_{Ac} = s_{AC}$. Therefore $\sum_c df_{Ac} = df_{AC}$, and likewise df_{AB} is the same in both economies; the same argument applies to B.

Step 4: intra-bloc trade cancels. Summing Eq. (3.3) over bystanders,

$$\sum_c \mathrm{TB}_c = \sum_c (f_{Ac} + f_{Bc}) - \sum_c (f_{cA} + f_{cB}) + \Big[\sum_c \sum_{c' \neq c} f_{c'c} - \sum_c \sum_{c' \neq c} f_{cc'} \Big],$$

and the bracket is identically zero. So $\sum_c \mathrm{TB}_c$ has the form of the composite's $\mathrm{TB}_C = f_{AC} + f_{BC} - f_{CA} - f_{CB}$, and $\mathrm{TB}_B = f_{AB} + \sum_c f_{cB} - f_{BA} - \sum_c f_{Bc}$ has the form of the composite's TB_B. Given Step 3, it remains to show $\sum_c df_{cA} = df_{CA}$ and $\sum_c df_{cB} = df_{CB}$.

Step 5: the bloc's demand for A's and B's goods. Since $d\log f_{cA} = d\log s_{cA} + d\log \tilde{I}_c - d\tau_{cA}$ and all three terms are common across bystanders, $\sum_c df_{cA} = (N-2) f_{cA}\, d\log f_{cA} = f_{CA}\, d\log f_{cA}$; we need $d\log f_{cA} = d\log f_{CA}$. Incomes: $d\log \tilde{I}_c = d\nu_C + s_{cA} d\tau_{cA} + s_{cB} d\tau_{cB}$, which equals $d\log \tilde{I}_C$ because $s_{CA} = s_{cA}$ and $s_{CB} = s_{cB}$ (Step 2). Tariff wedges: $d\tau_{cA} = d\tau_{CA}$ by construction. Shares: measure prices in bystander currency, so that other bystanders' goods have $dp = 0$, and write $dp_{cA} = \frac{S+D}{2}$ and $dp_{cB} = \frac{S-D}{2}$, where $S \equiv dp_{cA} + dp_{cB}$ and $D \equiv dp_{cA} - dp_{cB}$. A bystander's import price index moves by $S/(N-1)$, since A and B are two of its $N-1$ equally weighted origins. Hence

$$\begin{aligned} d\log s_{cA} &= (1-\rho)\Big(\frac{S+D}{2} - \frac{S}{N-1}\Big) + (1-\eta)\alpha_D \frac{S}{N-1} \\ &= (1-\rho)\frac{D}{2} + \Big[(1-\rho)\frac{N-3}{2(N-1)} + (1-\eta)\frac{\alpha_D}{N-1}\Big] S. \end{aligned}$$

In the composite the import index moves by $S/2$, so $d\log s_{CA} = (1-\rho)\frac{D}{2} + (1-\eta_C) h_C \frac{S}{2}$. The D terms coincide for any (S, D), and the S terms coincide if and only if $(1-\eta_C)(N-1)h_C = (1-\rho)(N-3) + 2\alpha_D(1-\eta)$. Using $(N-1)h_C = N - 3 + 2\alpha_D$, this is Eq. (3.11). The argument for B's good is identical with D replaced by $-D$. The residual share on bloc goods matches as well, because shares sum to one in both economies.

Step 6: equilibrium and NEER. By Steps 2–5, the two equations $(d\mathrm{TB}_B, \sum_c d\mathrm{TB}_c) = 0$ of the N-country economy, evaluated at $d\nu_c = d\nu_C$, are identical to the composite's equations $(d\mathrm{TB}_B, d\mathrm{TB}_C) = 0$, including the tariff terms. The composite system is nonsingular (its determinant is computed below), so the two economies have the same $(d\nu_B, d\nu_C)$. The composite's trade weights are $w_{AB} = 1/(N-1)$ and $w_{AC} = (N-2)/(N-1)$, so its NEER response $\frac{1}{N-1} d\nu_B + \frac{N-2}{N-1} d\nu_C$ equals the N-country NEER at symmetric weights.

Step 7: the reduced system. By Lemma A.1, row B of $-J$ in the N-country economy is $c(N-1)$ on ν_B and $-c$ on each bystander's rate; imposing $d\nu_c = d\nu_C$ gives $(c(N-1), -c(N-2))$. Each bystander row has $-c$ on ν_B, $c(N-1)$ on its own rate and $-c$ on each of the other

$N-3$ bystanders' rates; summing the $N-2$ bystander rows and imposing $d\nu_c = d\nu_C$ gives $(-c(N-2),\ (N-2)[c(N-1)-c(N-3)]) = (-c(N-2), 2c(N-2))$. This is Eq. (3.12), with determinant $c^2[2(N-1)(N-2)-(N-2)^2] = c^2N(N-2) > 0$. Inverting and applying it to $(F_B, (N-2)F_c)$ gives Eq. (3.13), and the unilateral disturbances follow from Eq. (A.3). □

The replication code verifies the reduction numerically: for $(\alpha_D, \alpha_T, \eta)$ equal to $(0.8, 0.4, 1.5)$, $(0.6, 0.7, 1.2)$, $(0.9, 0.3, 3.0)$ and $(0.5, 0.5, 2.0)$, $N \in \{4, \dots, 10\}$ and $\rho \in \{1.1, 4, 9\}$, under the four shocks (A on B; an A–B war; A on all bystanders; all bystanders on B), the composite's $(d\nu_B, d\nu_C)$ match the N-country responses to within 10^{-12}.

Proof of Proposition 3.5. Equations Eqs. (A.7) and (A.8) are derived in Section A. For (i), setting $d\log q_{AC} = 0$ and substituting $d\nu_C$ from Eq. (3.8) gives

$$\Big[(N-1) - mN\Big]\frac{\rho - N\rho_D}{N} = m(N-1)M_N(\rho).$$

Since M_N is affine in ρ, this is linear in ρ; for $m < 1/N$, solving gives $\rho_q^*(N)$. The coefficient $(N-1)-mN$ is then positive, and MML at ρ_q^* gives $M_N > 0$ on the right, so $\rho_q^*(N) > N\rho_D$. For (ii), under MML $M_N > 0$, and for $m < (N-1)/N$ both terms in Eq. (A.8) are negative. □

Proof of Proposition 4.1. (i) Let primes denote responses to the mirror shock $t_{BA} = d\tau$. Relabeling $A \leftrightarrow B$ in Eq. (3.8) gives

$$d\log e'_{BA} = -\frac{N\rho_D + (N-2)\rho}{N(N-1)M_N}\,d\tau, \qquad d\log e'_{BC} = \frac{\rho - N\rho_D}{N(N-1)M_N}\,d\tau.$$

Since $e_{AB} = 1/e_{BA}$ and $e_{AC} = e_{AB}e_{BC}$, $d\log e'_{AB} = \frac{N\rho_D+(N-2)\rho}{N(N-1)M_N}d\tau$ and $d\log e'_{AC} = \frac{\rho}{NM_N}d\tau$. By linearity the combined response is $d\log e_{AB} = 0$ and

$$d\log e_{AC} = \frac{\rho - N\rho_D}{N(N-1)M_N}\,d\tau + \frac{\rho}{NM_N}\,d\tau = \frac{\rho - \rho_D}{(N-1)M_N}\,d\tau.$$

The three components are A's average protection, $-\rho_D/[(N-1)M_N]$, A's relative treatment, $\rho/[N(N-1)M_N]$, and B's relative treatment, $\rho/(NM_N)$. By symmetry between A and B the same expression applies to $d\log e_{BC}$, every bystander moves identically, so cross rates among bystanders are unchanged, and averaging over A's $N-1$ partners gives $d\nu_A^E$ in Eq. (4.1).

(ii) Superpose the N tariff vectors. A's tariff vector is uniform, $\bar{t}^{(A)} = d\tau$ and $\tilde{t}^{(A)} = 0$, and contributes $\lambda_A d\tau$ to every $d\log e_{Aj}$. Partner j taxes only A; its contribution to $d\log e_{Aj}$ is $-[\lambda_A+(N-2)\lambda_R]\frac{d\tau}{N-1}$, while each of the $N-2$ other partners k contributes $-\lambda_R d\tau = \frac{\rho}{NM_N}d\tau$,

computed as in part (i) from $e_{Aj} = e_{Ak}e_{kj}$. Summing,

$$d\log e_{Aj} = \lambda_A d\tau - \Big[\lambda_A + (N-2)\lambda_R\Big]\frac{d\tau}{N-1} - (N-2)\lambda_R d\tau = \frac{(N-2)(\rho-\rho_D)}{(N-1)M_N}\,d\tau.$$

The expression is the same for every j, so it equals the effective response, and all cross rates among A's partners are unchanged. Under MML the signs are those of $\rho - \rho_D$. □

Proof of Lemma 5.1. Utility is Cobb–Douglas over the tradable aggregate and the non-tradable, so indirect utility is $V_A = \kappa\, I_A/(P_{TA}^{\alpha_T} P_{NA}^{1-\alpha_T})$ for a constant κ, with I_A, P_{TA} and P_{NA} in A's currency. The marginal utility of income is V_A/I_A, so the change in welfare in units of income is

$$\frac{dV_A}{(\partial V_A/\partial I_A)\, I_A} = d\log I_A - \alpha_T\, d\log P_{TA} - (1-\alpha_T)\, d\log P_{NA}.$$

This is the envelope theorem: by Shephard's lemma, $\alpha_T\, d\log P_{TA} = \sum_j s_{Aj}\, d\log p_{Aj}$, the expenditure-share-weighted change in consumer prices. Under the producer-price normalization $w_A = A_{TA}$ and $P_{NA} = A_{TA}/A_{NA}$ are fixed in A's currency, so $d\log P_{NA} = 0$ and the domestic tradable's price does not move. Imported goods' consumer prices move by $d\log p_{Aj} = d\nu_j + d\log(1+\tau_{Aj}) = d\nu_j + d\tau_{Aj}$ at free trade. Income is labor income plus rebated revenue, $I_A = w_A L_A/(1-\sum_j \theta_{Aj} s_{Aj})$ with $\theta_{Aj} = \tau_{Aj}/(1+\tau_{Aj})$. At free trade $\theta_{Aj} = 0$ and $d\theta_{Aj} = d\tau_{Aj}$, so changes in realized shares drop out and $d\log I_A = \sum_j s_{Aj}\, d\tau_{Aj}$: the revenue gain equals the tariff change times the border value of imports. Combining,

$$\frac{dV_A}{(\partial V_A/\partial I_A)\, I_A} = \sum_j s_{Aj}\, d\tau_{Aj} - \sum_j s_{Aj}(d\nu_j + d\tau_{Aj}) = -\sum_j s_{Aj}\, d\nu_j = -m_A \sum_{j\neq A} b_{Aj}\, d\nu_j,$$

with $m_A = \sum_j s_{Aj} = \alpha_T(1-h_A)$ and $b_{Aj} = s_{Aj}/m_A$. The tariff terms cancel: at free trade the revenue raised exactly offsets the direct price increase to first order. Tariffs imposed by other countries enter A's prices and income only through $d\nu$, so the formula holds for any tariff vector. □

The replication code checks Lemma 5.1 against central finite differences of A's log indirect utility in the nonlinear model, at symmetric and asymmetric free-trade baselines ($N = 3, 4, 5$) and under unilateral, retaliatory, multilateral-war, foreign-only and random tariff shocks; the two agree to within 10^{-6}.

Proof of Proposition 5.2. Work with the linearized system Eq. (A.1) at a balanced free-trade baseline, with country-specific η_i, ρ_i and α_T^i, so that $s_{Aj} = \alpha_T^A(1-h_A)b_{Aj}$ and $m_A = \sum_{j\neq A} s_{Aj} = \alpha_T^A(1-h_A)$.

Step 1: only A's import flows are disturbed. Set $d\nu = 0$ and $d\tau_{Aj} = dt$ for all $j \neq A$, with all other tariff changes zero. For $i \neq A$, $d\tau_{ij} = 0$, so $dp_{ij} = 0$ for every j, $d\log P_{Mi} = 0$, $d\log s_{ij} = 0$ and $d\log \tilde{I}_i = 0$. Every flow f_{ij} with $i \neq A$ is therefore unchanged. In the trade balance of partner j, the only term that moves is its exports to A:

$$F_j = d\mathrm{TB}_j\Big|_{d\nu=0} = f_{Aj}\Big(d\log s_{Aj} + d\log \tilde{I}_A - dt\Big).$$

Step 2: the cross-origin margin drops out. In A, $dp_{Aj} = dt$ for every j, so $d\log P_{MA} = \sum_j b_{Aj} dt = dt$ and $dp_{Aj} - d\log P_{MA} = 0$ for every j. The term multiplying $(1-\rho_A)$ vanishes, and

$$d\log s_{Aj} = (1-\eta_A)h_A\,dt \quad \text{for every } j, \qquad d\log \tilde{I}_A = \sum_j s_{Aj}\,dt = m_A\,dt.$$

Step 3: the disturbance. Substituting,

$$F_j = f_{Aj}\Big[(1-\eta_A)h_A + \alpha_T^A(1-h_A) - 1\Big]dt = -f_{Aj}\Big[\eta_A h_A + (1-h_A)(1-\alpha_T^A)\Big]dt = -f_{Aj}\,\rho_D^A\,dt,$$

which is Eq. (5.4). Since $\eta_A > 0$, $0 < h_A \leq 1$ and $\alpha_T^A \leq 1$, $\rho_D^A > 0$, so $F_j \leq 0$ for every j, with $F_j < 0$ whenever $f_{Aj} > 0$.

Step 4: exchange rates. Under MML, $(-J)^{-1} \geq 0$, so by Eq. (2.4) $d\nu_i = \sum_j \vartheta_{ij} F_j \leq 0$ for every i: A appreciates weakly against every partner. If A imports from every partner, then $F_j < 0$ for all j; since $(-J)^{-1}$ is nonsingular and nonnegative, each of its rows has a strictly positive entry, so $d\nu_i < 0$ for every i. If instead J is irreducible, $-J$ is an irreducible nonsingular M-matrix and $(-J)^{-1} > 0$ entrywise, so $d\nu_i < 0$ for every i as long as $F \neq 0$, that is, as long as A imports at all.

Step 5: welfare. By Lemma 5.1, $dU_A/(U_A' I_A) = -m_A \sum_j b_{Aj} d\nu_j \geq 0$, and it is strictly positive whenever some $d\nu_j < 0$ with $b_{Aj} > 0$, which holds in either case of Step 4 because the b_{Aj} sum to one. Since A's aggregate terms of trade improve if and only if $\sum_j b_{Aj} d\nu_j < 0$, they improve. Finally, the response is linear in the tariff vector, so the welfare effect of the uniform tariff is $\sum_j g_j\,dt$ with g_j as in Eq. (5.3), and $\sum_j g_j > 0$. □

The replication code checks Eq. (5.4) against the linear system to 10^{-12} and against complex-step derivatives of the exact model's trade balances, and verifies the signs of F, $d\nu$ and the welfare change on 60 random asymmetric baselines with country-specific parameters ($N = 3, 4, 5$).

Proof of Proposition 5.4. Under MML $\det A(\rho) > 0$, so $R_{A|B}$ has the sign of $Q_{A|B}(\rho)$. (a) At the symmetric benchmark, Eq. (A.9) gives $\det A = 3(mM_3/2)^2$ and $R_{A|B} = d\nu_A^E/d\tau = -\rho_D/(2M_3)$, so $Q = -\frac{3}{8}m^2\rho_D M_3$, which is affine in ρ ($q_2 = 0$) and negative for all ρ under

MML. (b) If $q_2 > 0$ and $q_0 < 0$, the product of the roots is $q_0/q_2 < 0$, so there is exactly one positive root ρ^*, with $Q < 0$ on $(0, \rho^*)$ and $Q > 0$ beyond. (c) If $q_2 < 0$, $q_1 > 0$, $q_0 < 0$ and the discriminant is positive, both roots are real with positive sum $-q_1/q_2$ and positive product q_0/q_2, so both are positive, and the downward parabola is positive only between them. □

C Away from the Symmetric Free-Trade Benchmark

The results of Sections 2.4, 3 and 4 are local to the symmetric free-trade benchmark. This appendix examines four departures from it: a positive common external tariff, finite tariff changes, asymmetric primitives, and the trade-weighted effective rate. The representation Eq. (2.3) and the MML condition apply at any balanced baseline. What changes is the mapping from primitives to the eigenvalues of Proposition 2.4 and, once partner symmetry fails, whether the common and zero-sum directions remain eigen-directions at all. In every case the baseline equilibrium is recomputed from primitives rather than by perturbing shares, and MML holds at every point reported. Unless stated otherwise, numerical values use the calibration of Fig. 2, $\alpha_D = 0.7$, $\alpha_T = 0.6$ and $\eta = 1.5$ ($\rho_D = 1.17$); US-like values are reported where they are discussed in the main text.

C.1 A positive common external tariff

Suppose A levies a common tariff τ_0 on all $N-1$ partners, which impose no tariffs of their own. This baseline treats the partners alike, so partner symmetry holds and Proposition 2.4 applies. Relative to free trade, A's currency appreciates by the same amount against every partner, so A's goods are more expensive abroad and baseline shares differ from their free-trade values. We measure tariff changes as $t_j = d\log(1+\tau_{Aj})$, which coincides with $t_j = d\tau_{Aj}$ at free trade. Let $\theta \equiv \tau_0/(1+\tau_0)$ denote the tariff wedge, μ_A the import share of A's tradable expenditure, μ_P the import share of a partner's tradable expenditure, and ω the share of A's good in a partner's import bundle. The rebate channel enters through $\phi \equiv \alpha_T\mu_A/(1-\theta\alpha_T\mu_A)$, the response of A's income to its import expenditure. Trade between A and each partner is balanced at the baseline, so $X \equiv f_{Aj} = f_{jA}$ denotes this bilateral flow, and $Y \equiv f_{jk}$ the flow between any two partners. Linearizing at this baseline gives

$$g_S = -X\rho_D(\tau_0), \qquad g_D = -X\rho, \qquad j_S = XM_S, \qquad j_D = \Big(X + (N-1)Y\Big)M_D, \tag{C.1}$$

where

$$\begin{aligned}\rho_D(\tau_0) &= 1-\phi(1-\theta)+(\eta-1)(1-\mu_A)(1+\phi\theta),\\ M_S &= 1+(\eta-1)(1-\mu_A)(1+\phi\theta)+(\rho-1)(1-\omega)+(\eta-1)(1-\mu_P)\,\omega,\\ M_D &= 1+(\rho-1)\frac{N-3+\omega}{N-2}+(\eta-1)(1-\mu_P)\frac{N-1-\omega}{N-2}.\end{aligned}$$

At free trade $\theta = 0$, $\phi = m$, $\mu_A = \mu_P = 1-\alpha_D$ and $\omega = 1/(N-1)$, so $\rho_D(0) = \rho_D$ and $M_S = M_D = M_N$.

The eigenvalues in Eq. (C.1) differ in the trade flows on which they act. Tariff disturbances along both components fall only on partners' trade with A, so g_S and g_D are proportional to X. A common exchange-rate movement leaves cross rates among partners unchanged and also shifts only partners' trade with A, so j_S is proportional to X, and X cancels from $\lambda_A = -\rho_D(\tau_0)/M_S$. A relative movement also shifts trade among partners. Under a zero-sum movement v, partner j's rate moves by v_j against A and by $v_j - v_k$ against partner k; since $\sum_{k\neq j} v_k = -v_j$, its average movement against the other partners is $(N-1)/(N-2)$ times v_j. The base $X+(N-1)Y$ is therefore partner j's trade with A plus its trade with the other partners, the latter weighted by this factor. Define $x \equiv X/(X+(N-1)Y)$, A's share of the trade on which relative exchange-rate movements act. Then $\lambda_R = -x\rho/M_D$ and $j_D/j_S = (1/x)(M_D/M_S)$. At free trade $X = Y$, so $x = 1/N$ and $j_D = Nj_S$: the factor N in the bystander threshold is $1/x$ evaluated at the free-trade baseline.

For a further tariff on B, $t = dt\, e_B$, the bystander response is $d\nu_C = (\lambda_A - \lambda_R)\, dt/(N-1)$, so A depreciates against C if and only if

$$\rho > \rho^*(\tau_0) \equiv \frac{\rho_D(\tau_0)}{x}\,\frac{M_D}{M_S}. \tag{C.2}$$

The condition is implicit in ρ, because ω, μ_P, x and M_S depend on baseline shares that themselves depend on ρ; we compute it by root-finding with the baseline re-solved at each ρ. Since the response is linear in t, a marginal preferential cut for B, $t = -dt\, e_B$, worsens C's terms of trade under the same condition. Table 5 reports the threshold relative to its free-trade value.

The threshold rises with τ_0, mainly because x falls. The common appreciation makes A's goods more expensive, so A's share of each partner's imports falls. At $N = 3$ and $\tau_0 = 1$, ω falls from $1/2$ to 0.29 and x from $1/3$ to 0.17. Relative treatment then disturbs a smaller part of the trade on which relative exchange-rate movements act. A rise in M_S relative to M_D offsets part of this effect. The local threshold also understates what a discrete preferential cut requires. In the exact model at $N = 3$, removing the tariff on B from $\tau_0 = 0.5$ worsens

Table 5: Bystander threshold at a common external tariff, $\rho^*(\tau_0)/N\rho_D$

	$\tau_0 = 0.25$	$\tau_0 = 0.5$	$\tau_0 = 1$
$N = 3$	1.12	1.22	1.41
$N = 5$	1.23	1.49	2.07

Notes: $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$. Computed from Eq. (C.2) with the baseline re-solved at each ρ. At US-like parameters the ratios are 1.10, 1.19, 1.36 ($N = 3$) and 1.23, 1.48, 2.07 ($N = 5$).

C's terms of trade only if $\rho > 6.23$, compared with 4.28 for a marginal cut at the same baseline and $3\rho_D = 3.51$ at free trade.

C.2 Finite tariffs

The thresholds of Propositions 4.1 and 3.2 concern marginal tariffs at free trade. In one case the finite-tariff sign is exact. At $N = 2$ and $\eta = 1$, expenditure shares are fixed at their weights, and $I_A = w_A L_A/[1 - \frac{\tau}{1+\tau}m]$, so $\mathrm{TB}_A = 0$ becomes $e\, m\, w_B L_B = m I_A/(1+\tau)$:

$$e(\tau) = \frac{w_A L_A}{w_B L_B} \cdot \frac{1}{1 + (1-m)\tau}, \qquad \left.\frac{d \log e}{d\tau}\right|_{\tau=0} = -(1-m) = -\left.\frac{\rho_D}{M_2}\right|_{\eta=1}.$$

Thus $e(\tau)$ is decreasing in τ, and the sign of the linearized response is exact for all tariff levels in this case.

For finite tariffs at $N = 3$, starting from the symmetric free-trade benchmark, let $\rho^*(\tau)$ denote the value of ρ at which $\log e_{AC}(\tau) = 0$ in the exact model. Writing $\log e_{AC}(\tau) = a(\rho)\tau + \frac{1}{2} b_C \tau^2 + O(\tau^3)$, where a is the first-order slope and b_C the second derivative along the tariff path,

$$\rho^*(\tau) = \rho_0^* + \kappa\, \tau + O(\tau^2), \qquad \kappa = -\frac{b_C}{2a'(\rho_0^*)}.$$

For a unilateral tariff, $a = (\rho - 3\rho_D)/(6M_3)$ by Eq. (A.10), so $\kappa_{\text{uni}} = -3M_3 b_C$. For bilateral retaliation, $\tau_{AB} = \tau_{BA} = \tau$ and $a = (\rho - \rho_D)/(2M_3)$ by Eq. (4.1), so $\kappa_{\text{war}} = -M_3 b_C$. Both are evaluated at the symmetric benchmark, where b_C follows from a second-order expansion of the equilibrium conditions.

The unilateral threshold falls with the size of the tariff. On a grid over $\alpha_D \in [1/3, 1)$, $\alpha_T \in (0, 1]$ and $\eta \in [1, 10]$, the exact finite-difference slope $(\rho^*(0.01) - 3\rho_D)/0.01$ is negative at every point. At $\tau = 1$ the exact threshold is 0.76 times $3\rho_D$ at the calibration of this appendix and 0.72 times $3\rho_D$ at US-like parameters; at the 2025 peak rate $\tau = 1.45$ it is 2.70 at US-like parameters, against $3\rho_D = 3.96$. A larger tariff shrinks B's share of A's imports, so a further increase adds less average protection while continuing to divert expenditure

toward C. The retaliation threshold instead rises slightly, with

$$\kappa_{\text{war}} = \frac{1-\alpha_D}{4}\Big[\alpha_D(\eta-1)^2 + 3\alpha_D\alpha_T(\eta-1) + 2\alpha_T(1-m)\Big] > 0 \qquad (\eta \geq 1),$$

and at $\tau = 1$ the exact threshold is 1.07 times ρ_D (1.03 at US-like parameters).

These two subsections illustrate a common principle. The threshold depends on the share structure on which the tariff acts. A common tariff shrinks A's footprint in its partners' trade and raises the threshold, while a concentrated tariff shrinks the target's share of A's imports and lowers it.

C.3 Asymmetric primitives

At a general balanced baseline the coefficients of Eq. (A.1) are the baseline shares b_{ij}, the home shares h_i and the incomes y_i. The elasticity ρ enters the share responses only through $(1-\rho)$, so every entry of J and F is affine in ρ. Each entry of $\text{adj}(-J)$ is therefore a polynomial of degree at most $N-2$ in ρ, and each numerator $[\text{adj}(-J)F]_i$ a polynomial of degree at most $N-1$. Since $\det(-J) > 0$ under MML, a bilateral response can change sign only at a root of this numerator. Symmetry reduces the problem to the two coefficients of Proposition 2.4, yielding linear thresholds. At a general asymmetric baseline the $N = 3$ numerator is already quadratic in ρ:

$$\text{sign}\,\frac{d\log e_{AC}}{d\tau} = \text{sign}\left(c_2\rho^2 + c_1\rho + c_0\right), \qquad c_2 = \alpha_T^2\, b_{AB}b_{AC}b_{CA}b_{CB}(1-h_A)(1-h_C)y_A y_C. \tag{C.3}$$

The coefficients c_1 and c_0 depend on the full pattern of trade shares and country sizes. Provided MML holds, $c_2 > 0$ whenever A and C each import from both foreign origins, so at given shares the response is positive beyond the largest positive root, which equals $3\rho_D$ at the symmetric point.

The quadratic holds at given shares, however, and the shares themselves depend on ρ once primitives are held fixed. At an asymmetric baseline relative prices differ across origins, and a higher ρ also reallocates expenditure toward the cheapest origin. A large target, whose currency must depreciate to balance its trade, can then crowd C out of A's imports and prevent the reversal. With $L_B = 2.2$ and all other primitives symmetric, e_{AC} reverses only for $5.99 < \rho < 53.8$, as A's import share from B rises from 0.74 at $\rho = 5$ to 0.997 at $\rho = 100$; with $L_B \geq 2.6$ it does not reverse for any $\rho \leq 1000$. Away from symmetry there may therefore be no bilateral threshold, one, or several, and for $N \geq 4$ the numerator is of higher degree in ρ.

The threshold also moves with asymmetric primitives near the symmetric benchmark.

Table 6 reports $d\rho^*/dx$ at $\rho = 3\rho_D$, where ρ^* solves $d \log e_{AC}/d\tau = 0$ after perturbing a single primitive x and recomputing the free-trade baseline. Derivatives are evaluated at $L_i = 1$ and $\beta_{AB} = 1/2$. Here β_{AB} is A's weight on B, with $\beta_{AC} = 1 - \beta_{AB}$, and α_D^A is the home weight in A alone.

Table 6: Sensitivity of the bystander threshold to asymmetric primitives, $N = 3$

	$d\rho^*/dx$	
x	$(\alpha_D, \alpha_T, \eta) = (0.7, 0.6, 1.5)$	US-like $(0.8, 0.4, 1.5)$
β_{AB}	+3.70	+4.06
L_B	+0.99	+1.23
L_C	−0.20	−0.32
α_D^A	+5.31	+6.61

The interpretations are as follows. β_{AB}*:* a tariff on a larger supplier is a larger average-protection shock, so B's loss rises relative to C's gain. L_B*:* a larger target depreciates to balance its trade, which raises A's import share from B. L_C*:* a larger bystander depreciates and gains import share in A, but also absorbs its own disturbance more easily; the two effects nearly offset. α_D^A*:* stronger home bias in A raises domestic diversion and makes C's rate respond more to B's loss.

C.4 The effective rate and the ordering of bilateral responses

At the symmetric benchmark, relative treatment has no effect on the equally weighted average of A's bilateral responses, and this average coincides with the trade-weighted NEER. A unilateral tariff therefore always appreciates A's effective exchange rate. Relative treatment drops out of the effective rate only when the trade weights and the adjustment structure are symmetric. With asymmetric trade weights, $d\nu_A^E = w_{AB}\, d\nu_B + w_{AC}\, d\nu_C$, and a bystander depreciation $d\nu_C > 0$ enters with weight w_{AC}. The trade-weighted NEER can then depreciate when the bystander reversal occurs and the bystander carries most of A's trade weight. Table 3 illustrates this at US-like parameters. The reversal starts below $3\rho_D$ because a small target with a small import weight lowers the threshold, as in Table 6. The exact model at $\tau = 0.01$ confirms the signs.

The ordering $d\nu_B < d\nu_C$, which holds at the symmetric benchmark because $\lambda_R < 0$, can also fail. This requires ρ near one and a high η, and $\rho < \eta$ in every case we examined. For example, with $L = (5.12, 4.26, 2.28)$ for (A, B, C), $\beta_{AB} = 0.86$, $\beta_{BA} = 0.65$, $\beta_{CA} = 0.19$, $\alpha_D = 0.89$, $\alpha_T = 0.6$, $\eta = 3.74$ and $\rho = 1.20$, $d\nu_B - d\nu_C = 0.08\, d\tau$ (MML holds; the exact model at $\tau = 0.01$ gives 0.079).

Online Appendix

D Further Robustness

D.1 Robustness of the two-country result

This subsection collects the monotonicity results behind Remark 2.1. Equilibrium is balanced trade for A, $\mathrm{TB}_A(e,\tau) \equiv X_A(e,\tau) - M_A(e,\tau) = 0$, where e is units of A's currency per unit of B's, τ is A's ad valorem tariff on imports from B, and X_A and M_A are the values of exports and imports in A's currency. Assume TB_A is continuously differentiable with $\mathrm{TB}_{A,e} \neq 0$ locally. By the implicit function theorem there is a locally unique mapping $e(\tau)$ with

$$\frac{de}{d\tau} = -\frac{\mathrm{TB}_{A,\tau}(e(\tau),\tau)}{\mathrm{TB}_{A,e}(e(\tau),\tau)}.$$

Under the producer-price normalization the consumer price of imports in A is $p_M(e,\tau) = (1+\tau)e$ and the price of A's exports in B is $p_X(e) = 1/e$.

Lemma D.1 (Tariff monotonicity)**.** *Fix e. Suppose (i) $p_M(e,\tau)$ is increasing in τ, and (ii) import expenditure $M_A(e,\tau)$ is decreasing in the consumer import price. Suppose in addition that any induced effect of τ on export revenue satisfies $X_{A,\tau} \geq 0$, or the weaker condition $X_{A,\tau} > M_{A,\tau}$. Then $\mathrm{TB}_{A,\tau} > 0$.*

Proof. $\mathrm{TB}_{A,\tau} = X_{A,\tau} - M_{A,\tau}$, and by (i)–(ii) $M_{A,\tau} = (\partial M_A/\partial p_M)(\partial p_M/\partial \tau) < 0$. Either condition on $X_{A,\tau}$ then gives $\mathrm{TB}_{A,\tau} > 0$. □

In the baseline model the tariff applies only to imports and there is no policy response abroad, so $X_{A,\tau} = 0$ at fixed e and the condition holds trivially.

Lemma D.2 (Exchange-rate monotonicity)**.** *Fix τ. If $M_{A,e} < 0$ and $X_{A,e} \geq 0$, with at least one inequality strict, then $\mathrm{TB}_{A,e} > 0$.*

Proof. $\mathrm{TB}_{A,e} = X_{A,e} - M_{A,e} > 0$ under the stated conditions. □

Proposition D.3 (Sign of the two-country response)**.** *If the conditions of Lemmas D.1 and D.2 hold in a neighborhood of an equilibrium (e^*,τ), then $e(\tau)$ exists locally and $de/d\tau < 0$.*

Proof. By the implicit function theorem and Lemmas D.1 and D.2, $de/d\tau = -\mathrm{TB}_{A,\tau}/\mathrm{TB}_{A,e} < 0$. □

The condition of Lemma D.2 is the Marshall–Lerner condition, and that of Lemma D.1 is its tariff analogue: import expenditure falls as the consumer import price rises. Both hold across standard specifications.

Demand. Tariffs and exchange rates affect demand only through relative prices. Under CES (Armington) preferences over tradables with any elasticity $\sigma > 0$, separable from non-tradables, Marshallian import demand is strictly decreasing in p_M, so at fixed e a higher tariff lowers import expenditure while export revenue is locally unaffected; holding τ fixed, a depreciation raises the domestic-currency price of imports and lowers the foreign-currency price of exports. The same holds under nested CES with separate elasticities between tradables and non-tradables and across tradable varieties: tariff-induced income effects may reallocate expenditure between tradables and non-tradables, but the monotone relation between import demand and its consumer price is preserved. Under quasi-linear preferences income effects from tariff revenue vanish and the conditions hold by construction.

Production. With diminishing returns or sector-specific factors, wages and output adjust to demand. At fixed e a higher tariff still raises the consumer price of imports and reduces import expenditure, and since the tariff introduces no export-side wedge, export revenue is not adversely affected. Holding the tariff fixed, a depreciation raises import prices and lowers export prices; provided export supply is weakly increasing in the export price, import expenditure falls and export revenue weakly rises, so $\mathrm{TB}_{A,e} > 0$ continues to hold locally.

Fiscal closure and pricing. Whether tariff revenue is rebated lump-sum, wasted, transferred abroad or used to finance government spending, a tariff raises the consumer price of imports at a fixed exchange rate, so the conditions of Lemmas D.1 and D.2 hold under standard fiscal closures. Local- or dominant-currency pricing and monopolistic competition mainly affect the degree and timing of pass-through; in the flexible-price equilibrium that our analysis describes, consumer prices reflect both the tariff wedge and the exchange rate, and the monotonicity conditions remain valid. While the magnitude of the response varies across specifications, its sign does not, wherever Marshall–Lerner holds.

D.2 Sector-specific factors

The linear technology makes the supply of tradables infinitely elastic at the going wage. To give the supply side curvature, replace $Y_{Ti} = A_{Ti}L_{Ti}$ with $Y_{Ti} = A_{Ti}L_{Ti}^{\gamma}$, $\gamma \leq 1$, as in the Ricardo–Viner model, with non-tradables linear as before and labor mobile across sectors within each country, $L_{Ti} + L_{Ni} = L_i$. Under the producer-price normalization the tradable's

price is still one in its own currency, so the exchange rate is still the relative price of tradables. Competitive pricing gives $w_i = \gamma A_{Ti} L_{Ti}^{\gamma-1}$ and $P_{Ni} = w_i / A_{Ni}$, and the fixed factor earns rents $(1-\gamma)Y_{Ti}$ that are rebated lump-sum, so income is $Y_{Ti} + P_{Ni} Y_{Ni}$ plus rebated tariff revenue. Equilibrium adds non-tradable market clearing, $w_i L_{Ni} = (1-\alpha_T) I_i$, in every country to the trade-balance conditions. With $\gamma = 1$ the model is the baseline, and the replication code confirms that it reproduces the baseline allocations and responses to 10^{-12}; its linearization matches finite differences of the exact model for $\gamma < 1$.

With Cobb–Douglas preferences, tradable employment at a balanced free-trade baseline is $L_{Ti} = \gamma\alpha_T L_i / (\gamma\alpha_T + 1 - \alpha_T)$, whatever the exchange rates. At the margin, therefore, neither the bystander's nor anyone else's supply of tradables responds to exchange rates, and the adjustment matrix J is only rescaled by nominal income. What changes is the tariff-setter's own income response. Rebated revenue raises A's demand for non-tradables and draws labor out of tradables, and with diminishing returns this raises A's wage and non-tradable price. At given exchange rates A's nominal spending therefore rises by more (6.25 rather than 4 percent per unit of $d\log(1+\tau_{AB})$ at $\gamma = 0.5$ and US-like parameters), and part of the extra spending falls on the bystander's good. The effect works like a smaller domestic diversion margin, so **diminishing returns lower the bystander threshold**, but only modestly at the symmetric point: from 3.96 to 3.88 at $\gamma = 2/3$ and to 3.83 at $\gamma = 0.5$. The effect is larger when the bystander is small. With $L_C = 0.1$ and all other primitives symmetric, the threshold is 4.98 with linear technology, 4.43 at $\gamma = 2/3$ and 4.19 at $\gamma = 0.5$. In every case there is a single threshold for $\rho \in [1.05, 60]$, MML holds at it, and A appreciates against the target there. None of the sign results of the paper requires diminishing returns; the linear baseline delivers them.

D.3 Permanent trade imbalances

Balanced trade is a modelling choice, not a requirement of the representation Eq. (2.3), which applies at any baseline. Suppose A receives a permanent transfer T_A, fixed in A's currency and paid by one partner, so that A's spending exceeds its income by T_A and equilibrium requires $\mathrm{TB}_i + T_i = 0$ for every i. Tariff revenue is rebated and trade is measured at border prices as before, and with $T = 0$ the model is the baseline (again confirmed to 10^{-12}). We set T_A to 3 percent of A's income, a US-scale deficit, at US-like parameters with $N = 3$ and otherwise symmetric primitives. Because A's import share of income is only 0.08, the deficit is large relative to trade, and financing it requires a baseline appreciation of A of about 15 percent against the paying country and 8 percent against the third.

The threshold structure is unchanged, but the deficit raises the bystander threshold from

3.96 to 4.94 when the surplus counterpart is the target B and to 4.61 when it is the bystander C. When B pays, A's imports tilt toward the cheapened target: C's share of A's imports falls from 0.50 to 0.45, and C's sales to A fall from one-half to 0.39 of its exports, so the bystander is less exposed to diversion from A's tariff. When C pays, the bystander is poorer and smaller in A's currency, which raises the threshold as a small bystander does in Section D.2. The effect is roughly linear in the imbalance: a deficit of 1 percent gives 4.21 (B pays) and 4.14 (C pays), and a surplus of 3 percent lowers the threshold to 3.46 and 3.56.

At 2025 tariff levels the size of the tariff dominates. For a discrete tariff $\tau_{AB} = 1.45$, the nominal bystander threshold is 2.70 at a balanced baseline (Section C.2) and 2.94–3.07 at the 3 percent deficit. For the real exchange rate q_{AC} the marginal threshold is higher, 4.93 at a balanced baseline (Proposition 3.5) and 5.64–6.33 with the deficit, but the discrete tariff again lowers it, to 3.19 at a balanced baseline and 3.42 (C pays) to 3.64 (B pays) with the deficit. Combining the real rate, the deficit and a tariff of 2025 size thus leaves the threshold below the free-trade nominal benchmark $3\rho_D = 3.96$.